Charged CL-FTS: theory, validation and exact sampling controls

Consolidated results, code map and execution protocols

2026-09-08; numerical campaigns through 2026-09-07

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1 Scope and current conclusion

This document consolidates the charged-polymer theory notes, implementation checks, native complex Langevin (CL) validation, exact finite-action benchmarks, and positive-sampling controls. PRL-specific N=100 reproduction and its planar MC comparison are in the separate PRL report.

Current verdict: the native charged CL sample means disagree with exact references in the investigated finite model. Independent chain/drift/noise checks pass, and positive representations of the same action recover reference means. This supports incorrect CL convergence in that fixture; it is not a proof that all CL fails or a complete boundary-term derivation. A drift-tail study (2026-10-26) finds the power-law tail of a partition-function zero in every setting of this fixture. The bias grows with the weight of that tail, and the CL averages agree with the exact sums where

it is light: without charges at $\zeta N \leq 1$, and up to $E = 10$ at $\zeta N = 0.25$–0.5. The FFT strong-field issue is separately documented and remains deferred by user decision.

Historical weak-coupling, one-loop and brush observations are retained later in this document with their original parameters and uncertainties. They do not supersede the stronger exact-reference checks or certify native CL for all models.

2 Code used and current organization

Directory organization updated 2026-09-14: models are in `models/`, tests in `tests/`, and this report and its build script in `reports/`. All code paths in prose are relative to `devel/charged_polymers`; `workflow.py` commands are unchanged.

Historical command names are kept in `studies/legacy_commands.json`; the current entry point is `workflow.py`. Model modules are organized under `models/`. The source revision immediately before consolidation is d395ed1a.

Purpose	Historical script(s)	Current command / implementation
Deterministic implementation tests	`validate_clfts.py`	`workflow.py check implementation`; `studies/validation.py`
Same-model MF/CL brush	`validate_brush_sampling.py`	`workflow.py cl brush`; `studies/validation.py`
Exact finite particles/native CL	`validate_clfts_exact.py`	`workflow.py cl exact`; `studies/exact.py`
Reduced CL and alternative integrators	`diagnose_clfts_reduced.py`	`workflow.py cl reduced`; `studies/reduced.py`
Final checkpoint density check	`check_clfts_exact_checkpoint.py`	`workflow.py check checkpoints`; `studies/cl_diagnostics.py`
Original field integral	`check_clfts_field_integral.py`	`workflow.py check field-integral`; `studies/cl_diagnostics.py`
Partition zeros and drift extremes	`check_clfts_poles.py`	`workflow.py check poles`; `studies/cl_diagnostics.py`
Stationarity and cutoff contributions	`analyze_clfts_identities.py`	`workflow.py check identities`; `studies/cl_diagnostics.py`
One-real-field positive marginal	`validate_clfts_positive_marginal.py`	`workflow.py sample marginal`; `studies/marginal_sampling.py`
4/16/32-cell Gibbs sampler	`sample_positive_1d.py`	`workflow.py sample finite`; `studies/finite_sampling.py`
Native/reduced/MF/marginal summaries	`analyze_clfts_exact.py`, `analyze_clfts_diagnosis.py`, `summarize_validation.py`, `summarize_clfts_pole_resolution.py`	`workflow.py analyze cl-exact`, `cl-reduced`, `cl-brush`, `marginal`; `studies/cl_analysis.py`

The common autocorrelation and block-error estimators are in `studies/statistics.py`; their numerical definitions are unchanged. `models/clfts_charged.py`, `models/brush_clfts.py`, `models/electrostatics.py`, and `models/stable_grafted_chain.py` retain the actual CL, wall, electrostatic and reference chain implementations. `models/brush_scft_charged.py` retains the historical mean-field brush solver, including its own Poisson saddle. The earlier charged L-FTS prototype and its model-specific neutral-reduction/Debye–Hückel scripts were retired; the charged CL neutral-reduction check remains. Distinct exact sums, field quadrature, direct chains, and log-chain references are intentionally not collapsed into one implementation: their independence is part of the validation.

`models/common.py` holds the parameter-serialization helper and per-species smearing adapter shared by the brush SCFT and CL classes. They were extracted on 2026-09-14 without changing their behavior. The relocation record retains the earlier hashes and includes the common module in current source audits.

3 Execution protocols

3.1 Environment and general acceptance rules

```
source /home/yongdd/anaconda3/etc/profile.d/conda.sh
conda activate polymerfts
ulimit -s unlimited
export OMP_STACKSIZE=1G OMP_NUM_THREADS=1
export MKL_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 OMP_MAX_ACTIVE_LEVELS=0
export PYTHONPATH=/home/yongdd/polymer/dh_salt_runs/exact_cl_diagnosis_20260907/python_deps
```

Use the installed `polymerfts` extension; rebuild/install first if changing its C++/CUDA sources. Numba/llvmlite for the reduced and positive samplers were loaded from the recorded local directory. Check `nvidia-smi` and select an idle GPU for CUDA. No production FFT/C++/CUDA kernels were changed during this consolidation.

Computations longer than one minute use SLURM, one job per parameter configuration. A job script contains this environment and one command; a typical CPU submission is `sbatch --cpus-per-task=1 --mem=8G --time=24:00:00 job.sh`, adjusted to local partition limits. All examples below are job bodies or short deterministic checks, not instructions to run long trajectories on a login node. Use new output paths.

Numerical tests retain calibrated grids, fields and tolerances. Random-field checks use standard deviation about 5. Check material conservation and `check_total_partition()` alongside final densities. A successful exit, finite trajectory, matching height alone, or a favorable drift-tail fit does not establish the correct CL measure.

3.2 Deterministic checks and the same-model MF/CL bridge

`workflow.py check implementation` runs strong-field references, force–Hamiltonian finite differences, annealed/quenched graft comparisons, CPU/CUDA density comparisons, checkpoint/restart and consistent-time monitoring, zero-charge reduction, and fixed-fugacity Donnan/PB checks. The eight production partition checks use relative complex error with the original 10^{-7} tolerance; FFT propagation itself remains unchanged.

The separate MF/CL bridge uses 2D 8×32, L=(1.2,2), N=10, E=25, $\zeta N = 20$, $\chi_{PS}N = 10$, five canonical species, C=1 and SP=3 valences, quenched grafts and truncated-bottom/reflecting-top bonds. Set `alpha_ds=0`. First solve all active mean-field equations including their uniform modes; only the electrostatic gauge is fixed. Then warm-start CL from that saddle.

- nbar=10^4, dt=.05/.025, seed 51: total time 500, discard 100.
- nbar=10^4, dt=.05/.025, seed 52: total time 2000, discard 100.
- nbar=10^8, dt=.05/.025, seed 51: total time 500, discard 100.
- Store profiles and observables every .25 Langevin units, use 20 blocks, and compare block and autocorrelation SE. Retain species profiles, drift and RNG state.
- A 16×64 MF refinement is a separate spatial check, not a retuned benchmark.

```
python devel/charged_polymers/workflow.py check implementation \
  --output /path/to/new/deterministic.json --platforms cpu-mkl cuda
python devel/charged_polymers/workflow.py cl brush \
  --dt .025 --nbar 1e4 --seed 52 --duration 2000 --equil 100 \
  --platform cpu-mkl --out /path/to/new/mf_cl_seed52
```

3.3 Exact finite particles versus native CL

The common fixture has one grafted four-bead P chain, two C(+1), SP(+3), SM(-1), and S: nine beads total, eight free coordinates, reflecting bonds, $L = 1$, $N_{\text{ref}} = 4$, $\chi N = 2$, $\zeta N = 4$, and Gaussian smearing radius 0.1. Every microscopic configuration of the same finite action is summed: $2^8 = 256$ or $4^8 = 65{,}536$. Uniform Gaussian modes cancel in normalized observables; self terms and canonical multiplicities remain in the exact reference.

The initial 22 native runs use two cells with E=0/25/100, dt=.02/.01/.005, stabilization off, seeds 71/72; two-cell E=25, dt=.01 with stabilization .05; and four-cell E=25, dt=.01 with stabilization off. Each lasts 2000 Langevin units, discarding 200, recording every .1. Two further two-cell

E=25/100 runs use dt=.0025, seed 173, total time 1000, discard 100. These give 24 completed runs, with different physical durations explicitly retained.

The original ψ stream uses seed+1: seeds 71/72 overlap across different field components and are reported separately. They are not pooled as fully independent replicates. Seed 173 provides a disjoint stream assignment.

```
python devel/charged_polymers/workflow.py cl exact \
  --sites 2 --coupling 100 --dt .0025 --duration 1000 --equil 100 \
  --seed 173 --stabilization 0 --out /path/to/new/exact_native
python devel/charged_polymers/workflow.py analyze cl-exact /path/to/campaign
python devel/charged_polymers/workflow.py check checkpoints /path/to/campaign
```

Compare height and all species/cell densities, real and imaginary means, 10/20/40-block errors, autocorrelation, ESS, mass and contour consistency. The reported SE is the maximum of block and autocorrelation estimates. Record $|Q(W)|/Q(\mathrm{Re}\,W)$ rather than treating a small raw Q as a pole. The checkpoint test compares final conditional densities independently; it is not a whole-trajectory or maximum-drift validation.

3.4 Reduced CL, direct field integration, and poles

The two-cell equations are reduced exactly to four complex amplitudes without changing the action. Check native/reduced densities, drifts, noise variance and one-step updates before running alternate integrators. The 21-trajectory protocol is listed in full in the diagnosis chapter: LM/ETD, Heun, mobility controls and adaptive Euler, steps down to .00015625, physical durations up to 50,000, and explicit equilibration intervals.

Adaptive Euler chooses the step from the current drift before drawing noise: $dt_{\text{actual}} \le dt_{\max} \min(1,\, 10/\max|\text{drift}|)$. Noise scales with its square root. Samples remain spaced by .1 in physical time, not equally weighted by adaptive step count. No rejection, clipping, or stabilizing force is added.

```
python devel/charged_polymers/workflow.py cl reduced \
  --coupling 100 --method heun --dt .00015625 --duration 20000 \
  --equil 2000 --seed 501 --psi-mobility 1 \
  --out /path/to/new/reduced_heun
python devel/charged_polymers/workflow.py check field-integral \
  --setup /path/to/saved/setup.npz --order 32 \
  --out /path/to/new/quadrature32.json
python devel/charged_polymers/workflow.py check poles /path/to/reduced_campaign
```

The direct field integral evaluates the entire product of partition functions on the original contour, using tensor Gauss–Hermite quadrature and no logarithm or division by Q. Compare increasing quadrature orders and exact particle sums. Pole analysis locates analytical chain/ion zeros and recomputes native densities at saved worst-drift states. Generator identities and cutoff contributions use the original total sample count; deleting near-pole samples and conditionally renormalizing would change the estimator and is not used.

3.5 Same-action positive marginal and 1D Gibbs control

The one-real-field marginal integrates repulsive and electrostatic Gaussian fields exactly and sums the 256 particle configurations conditionally. Use E=0/25/100, real Heun dt=.005/.00125, seeds 901/1001, total time 20,000, discard 2,000 and save every .1, yielding 180,000 samples per run. All 19 observables are checked; real quadrature and a std=5 finite-difference force check validate the marginal. This is an exact change of representation, not native CL dynamics.

The scalable-to-more-cells Gibbs control retains particles plus the attractive real field rather than summing all particle configurations at every field value. Use M=4/16/32, E=0/100, joint seeds 1101/1201, particle-only seed 2101; add M=4,E=25 with the same three choices. Each of the 21 runs has 2,000,000 sweeps, discards 200,000 and saves every 10. A sweep updates eight free particles and the conditional real field. Check exact sums at M=4 and compare the two positive representations at larger M; larger grids have no exhaustive reference.

```
python devel/charged_polymers/workflow.py sample marginal \
  --setup /path/to/saved/setup.npz --dt .00125 --seed 901 \
```

```
  --duration 20000 --equil 2000 --out /path/to/new/marginal
python devel/charged_polymers/workflow.py sample finite \
  --sites 16 --coupling 100 --mode joint --seed 1101 \
  --field-setup /path/to/saved/setup.npz \
  --sweeps 2000000 --burn 200000 --stride 10 --out /path/to/new/gibbs
```

The supplied `setup.npz` must match the intended fixture and coupling. Use the actual setup path from the original run script, rather than an arbitrary warm field. The N=100 planar sampler has a different finite action and is documented with PRL comparisons in the other report; it is not substituted into this test.

4 Evidence, provenance, and statistical interpretation

Evidence roots are below `/home/yongdd/polymer/dh_salt_runs/`:

Directory	Evidence
`validation_20260906/`	Deterministic results, strong fields, same-model saddle/CL and original regression logs
`exact_cl_20260907/`	24 native CL runs, exact references, final checkpoints, `analysis.json`
`exact_cl_diagnosis_20260907/`	21 reduced trajectories, four quadrature jobs, pole checks, setup/RNG snapshots
`exact_cl_pole_resolution_20260907/`	12 positive marginals, six generator/cutoff analyses, failed diagnostic compile logs
`oned_prl_20260907/`	21 small-system Gibbs runs; PRL entries in this mixed directory belong to the other report

`reports/metadata/campaigns.json` preserves exact stored parameters and run-script hashes. The old documents are merged into `archive/RESEARCH_LOG.md` for provenance. To replay a historical command against the consolidated code, use `workflow.py legacy OLD_SCRIPT.py` with unchanged arguments and a fresh output path. For exact source replay, use the archived code snapshot or recorded Git revision. Native RNG streams, action parameters and production FFT are unchanged.

Correlated observables and paired seeds are not independent hypothesis tests. Empirical SE, a small imaginary mean, or low-order stationarity alone cannot certify CL. The result chapters below retain failures and limitations alongside successful deterministic checks. The cleanup itself is a code/document refactor; it is not a new physical sampling campaign.

5 Charged field theory: action, units and discretization

> Working notes (devel). Formulation of polyelectrolyte solutions within this codebase's multi-monomer field theory, following the smeared-charge model of Wang [1] and Villet–Delaney–Fredrickson [2,3], written in the conventions of *Macromolecules* **2025**, 58, 816 (the neutral multi-monomer theory this library implements). **Compressible (finite ζN) model with per-species density smearing** — the fully UV-regularized formulation.

5.1 1. Model

Species $i = 1, \dots, S$ (here: polymer P, solvent S, counter-ion C, salt ions SP/SM). Each species is a (possibly very short) bead-spring chain of the discrete or continuous model already implemented.

Smeared densities. Every species carries a normalized Gaussian shape function

$$h_i(\mathbf{r}) = \frac{1}{(2\pi a_i^2)^{3/2}} e^{-r^2/2a_i^2}, \qquad \hat{h}_i(\mathbf{k}) = e^{-a_i^2 k^2/2},$$

of radius a_i (params `radiuses`, in R_0 units). a_i is the **smearing length** (Fredrickson school: width of the "shape function" or form factor of the species [2,3]); for charged species Wang [1] identifies it with the ion's **Born radius**, since the Gaussian self-energy $u_{\text{self}} = z_i^2 l_B/(2\sqrt{\pi}\, a_i)$ is the Born solvation energy of an ion of that size. ALL interactions are written in terms of the smeared volume fractions

$$\bar{\phi}_i \equiv h_i * \phi_i.$$

Smearing regularizes both the contact (excluded-volume/χ) and the Coulomb self-interactions, making the fluctuating field theory UV-convergent [1,2,3]. The point-particle limit $a_i \to 0$ is singular for a compressible fluctuating model and must not be taken. (`$a_i = $ None` ⇒ $h_i = \delta$; acceptable only for testing or in mean-field.)

Three interaction channels:

1. **Flory–Huggins**: χN_{ij} pairs (here only P,S — all pairs involving ions are zero). No separate `chi_monomers` machinery is needed in the compressible model: the χN matrix is simply $S \times S$ with zero rows/columns for the ions (see §3).
2. **Compressibility (Helfand)**: instead of the incompressibility constraint, a finite penalty

   $$\beta U_\zeta = \frac{\rho_0 \zeta N}{2N} \int \Big(\sum_i \bar{\phi}_i(\mathbf{r}) - 1 \Big)^2 d\mathbf{r}$$

   with the code's existing `zeta_n` parameter. The incompressible theory is the $\zeta N \to \infty$ limit. In a fluctuating simulation the smeared $\bar{\phi}_i$ (not raw ϕ_i) must enter here, or the contact penalty is UV-divergent.
3. **Coulomb**: segment of species i carries per-segment valence z_i; the smeared charge density is

   $$c(\mathbf{r}) = \sum_i z_i\, \bar{\phi}_i(\mathbf{r}) \qquad (\text{same } h_i \text{ as above}),$$

   and

   $$\beta U_C = \frac{l_B \rho_0^2}{2} \iint \frac{c(\mathbf{r})\, c(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, d\mathbf{r}\, d\mathbf{r}', \qquad l_B = \frac{e^2}{4\pi\varepsilon k_B T}$$

   (Bjerrum length; uniform dielectric ε, no dielectric contrast in this first version).

Global electroneutrality is still required (compressibility does not fix charge). With per-segment valences z_i and equal segment volumes:

$$\sum_i z_i\, \bar{\phi}_i^{\text{tot}} = 0$$

over overall volume fractions (NO division by chain length — that would be per-chain counting, wrong for per-segment valences). The counter-ion fraction is NOT a free parameter: for P with $z_P = +1$ and fraction ϕ_P, one needs $\phi_C = \phi_P z_P/|z_C|$ (plus balanced salt $z_{SP}\phi_{SP} + z_{SM}\phi_{SM} = 0$). The current `examples/Lamella.py` computes the counter-ion fraction from this condition. Its defaults $z_P = 0.2$, $\phi_P = 0.1$, and $z_C = -1$ give $\phi_C = 0.02$ and $\phi_S = 0.88$, so the input composition is electrically neutral.

5.2 2. Units and code conventions

Same as the neutral theory: lengths in $R_0 = a_{\text{Ref}} N_{\text{Ref}}^{1/2}$, fields per **reference chain** ($w_{\text{code}} = N\, w_{\text{per segment}}$), densities in ρ_0. Chain number density $C = \rho_0 R_0^3/N = \sqrt{\bar{N}}$.

Nondimensionalize with $\mathbf{x} = \mathbf{r}/R_0$. The Coulomb energy in k_BT becomes (note the C^2 — Coulomb is quadratic in ρ_0, unlike the χ/ζ block where $\chi N, \zeta N \propto \rho_0$ leave one C):

$$\beta U_C = \frac{C^2 E_0}{8\pi} \iint \frac{c(\mathbf{x})\, c(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}\, d\mathbf{x}\, d\mathbf{x}' = \frac{C^2 E_0}{2} \sum_{\mathbf{k}} \frac{|\hat{c}_{\mathbf{k}}|^2}{k^2}, \qquad E_0 \equiv \frac{4\pi l_B N^2}{R_0}.$$

For the per-chain-normalized Hamiltonian used throughout this codebase, the convenient coupling is

$$\boxed{E \;\equiv\; C\,E_0 \;=\; \frac{4\pi l_B N^2 C}{R_0} \;=\; 4\pi l_B\,\rho_0 N R_0^2}$$

(density-DEPENDENT: $E = \sqrt{\bar{N}}\,E_0$). Caveat: the literature's "electrostatic strength" [2,3] is the density-independent E_0, usually in $R_g = R_0/\sqrt{6}$ units — convert when comparing. New scalar param: `"bjerrum_e"` $= E$ (or `"bjerrum_length"` $= l_B/R_0$ with E computed internally from `nbar`).

5.3 3. Field theory: Hubbard–Stratonovich structure

Two independent HS blocks (the compressible model has NO separate pressure constraint):

(a) χ+ζ block — the existing compressible SPT, over ALL S species. The quadratic form of channels 1+2 is

$$\tfrac{1}{2}\,\bar{\phi}^{\mathsf{T}}(\zeta N\,\mathbf{J} + \chi N)\bar{\phi} \;-\; \zeta N\,\mathbf{1}^{\mathsf{T}}\bar{\phi} + \tfrac{1}{2}\zeta N,$$

with **J** the all-ones matrix — exactly the matrix $u = \zeta N\,\mathbf{J} + \chi N$ that `SymmetricPolymerTheory`'s compressible branch (`zeta_n` set) already diagonalizes, including the linear term (its `vector_large_s`). Ion rows of χN are zero. Spectrum for the 5-species example: - one large positive eigenvalue $\approx S\zeta N$ (the "pressure-like" mode) — imaginary-type; - the χ-split P–S pair: one negative (exchange, **real** Langevin field), one positive (imaginary-type); - pure ion-composition modes orthogonal to **1** with no χ: eigenvalue **0** — ideal mixing in those channels; the corresponding auxiliary fields are identically zero and SPT's existing zero-eigenvalue handling (Gram–Schmidt + warning) applies. This is physical, not an error: those composition fluctuations are free.

Because $\zeta N\,\mathbf{J}$ lifts the would-be singularity of the χ-only matrix, **the full S-species SPT can be used directly** — the chi-subset wrapper of the incompressible draft is no longer needed.

(b) Coulomb block — new. HS on the positive-definite Coulomb quadratic form introduces the electrostatic potential $\psi(\mathbf{x})$:

$$e^{-\beta U_C} = \int \mathcal{D}\psi\;\exp\!\Big[-C\int d\mathbf{x}\,\frac{\nabla\psi\cdot\nabla\psi}{2E} \;+\; i\,C\int d\mathbf{x}\;c(\mathbf{x})\,\psi(\mathbf{x})\Big]\Big/\mathcal{N},$$

which reproduces $\exp[-\frac{C^2 E_0}{2}\sum_k |\hat{c}_k|^2/k^2]$ exactly because $E = CE_0$ (Gaussian integral over ψ). ψ is the per-reference-chain potential $N\beta e\varphi_{\text{phys}}$. Coupling with $+i$ to a real density and a positive-definite kernel ($4\pi l_B/k^2$, analogous to $\lambda > 0$): ψ is an **imaginary-type field**, partial-saddled like the positive-eigenvalue SPT fields — see §5.

5.4 4. Hamiltonian and single-chain problem

The per-chain effective Hamiltonian is

$$\frac{H[\{w_k\},\psi]}{C\,k_B T V/R_0^3} = \underbrace{h_{\text{const}} + \sum_k \left[A_k\langle w_k\rangle + B_k\langle w_k^2\rangle\right]}_{\text{neutral SPT part}} + \frac{1}{V}\int \frac{\nabla\psi\cdot\nabla\psi}{2E}d\mathbf{x} - \sum_p \frac{\bar{\phi}_p}{\alpha_p}\ln Q_p[\{W_i^{\text{HS}}\}],$$

with the h-coefficients of the existing compressible SPT. The one-body potential of each species picks up the smearing convolution on EVERY term (the interactions couple to $\bar{\phi}_i = h_i * \phi_i$, so the

functional derivative w.r.t. ϕ_i convolves back with h_i):

$$\boxed{W_i^{\mathsf{HS}}(\mathbf{x}) \;=\; h_i * \Big[W_i^{\mathsf{SPT}} \;-\; i\, z_i\, \psi\Big](\mathbf{x})}$$

where W_i^{SPT} is the usual `matrix_a` mapping of the auxiliary fields (now including the pressure-like ζ mode; ion rows couple only to that mode and to ψ). No explicit N multiplies the electrostatic term: ψ is already per-reference-chain and z_i is per-segment — the code's $w = N w_{\mathsf{seg}}$ convention is carried entirely by ψ. The smearing is a k-space multiplication $\widehat{h}_i(k)\,[\widehat{W_i^{\mathsf{SPT}}} - i z_i \widehat{\psi}](k)$ — cheap, per species, applied once per saddle iteration; the propagator solvers are untouched.

Rotated contour / stored code field. As with the positive-eigenvalue SPT modes, the code stores $\psi_{\mathsf{code}} = -i\psi$. It is real at the saddle and complex during CL sampling. The HS charge term above then becomes $+z_i\psi_{\mathsf{code}}$ in the propagator input. The explicit electrostatic action becomes

$$\frac{1}{V}\int \frac{\nabla\psi\cdot\nabla\psi}{2E}\,d\mathbf{x} \;\longrightarrow\; -\frac{1}{V}\int \frac{\nabla\psi_{\mathsf{code}}\cdot\nabla\psi_{\mathsf{code}}}{2E}\,d\mathbf{x} \;=\; \frac{1}{2EV}\int \psi_{\mathsf{code}}\nabla^2\psi_{\mathsf{code}}\,d\mathbf{x}.$$

This is an analytic continuation: no complex conjugate or modulus square is used off the saddle. Only at a Poisson-consistent saddle does the last term equal $-\frac{1}{2V}\int c\psi_{\mathsf{code}}\,d\mathbf{x}$. This mirrors the negated B_k coefficients of the imaginary SPT fields. Note the $+c\psi_{\mathsf{code}}$ coupling itself lives inside $-\ln Q$ via W_i^{HS}; the explicit term recorded in H is only the (negative) gradient piece.

Concentrations: standard ϕ_i from propagators; the smeared densities that enter all interaction terms and the Poisson source are $\bar{\phi}_i = h_i * \phi_i$, $c = \sum_i z_i\bar{\phi}_i$ (again diagonal in k).

5.5 5. Forces and saddle conditions

- SPT auxiliary fields: unchanged from the neutral compressible theory — functional derivatives evaluated with the SMEARED densities $\bar{\phi}_i$ in place of ϕ_i. Negative-eigenvalue (exchange) fields receive Langevin noise; positive-eigenvalue fields are partial-saddled by the existing compressor loop; zero-eigenvalue fields stay zero.
- Electrostatic field:

  $$\frac{1}{C}\frac{\delta H}{\delta\psi_{\mathsf{code}}(\mathbf{x})} = \frac{\nabla^2\psi_{\mathsf{code}}}{E} \;+\; c(\mathbf{x}) \qquad\Longrightarrow\qquad \text{saddle: } -\nabla^2\psi^*_{\mathsf{code}} = E\,c(\mathbf{x}),$$

  the (smeared) **Poisson equation**; with the ideal-gas ions responding through their Boltzmann factors this is the fluctuating-field generalization of Poisson–Boltzmann. In k-space the solve is diagonal and exact:

  $$\hat{\psi}^*_{\mathsf{code}}(\mathbf{k}) = \frac{E\,\hat{c}(\mathbf{k})}{k^2}, \qquad k\neq 0,$$

  and the $k=0$ mode is fixed by electroneutrality ($\hat{c}(0)=0$ holds exactly per configuration in the canonical ensemble once §1's constraint holds; set $\hat{\psi}(0)=0$ as gauge). In the remaining update and discretization notes, ψ denotes the stored rotated field ψ_{code}.

Debye sanity check (1-segment ions, $z=\pm1$, ideal): linearizing gives $\kappa^2R_0^2 = E\,\bar{\phi}_{\mathsf{ion}}/N = 4\pi l_B\rho_0\bar{\phi}_{\mathsf{ion}}R_0^2$ — the physical Debye constant. Any factor error in E or W_i^{HS} shows up here first; keep this as a unit test.

CL-FTS (the real implementation): ψ fluctuates. L-FTS holds imaginary-type fields at partial saddle, so it can never capture ψ fluctuations — the charged theory belongs in CL-FTS, where ψ evolves with complex Langevin dynamics exactly like the pressure field W_+ (rotated storage,

imaginary-direction noise, "+Λ" drift):

$$\psi \;\leftarrow\; \psi + \left(\frac{\nabla^2\psi}{E} + c\right)\Delta t\, s_\psi \;+\; i\,\mathcal{N}(0,\sigma)\sqrt{s_\psi},$$

whose drift fixed point is the Poisson equation. Three discrete-scheme facts (all found the hard way): - The ∇^2/E part is stiff at high k: explicit Euler is UNSTABLE ($k^2\Delta t/E$ exceeds the stability limit), and semi-implicit Euler, while stable, SUPPRESSES the stationary variance by $1/(1+2\kappa\Delta t)$ per mode — this silently wrecks fluctuation thermodynamics (the one-loop $\langle\partial H/\partial E\rangle$ came out 2–$4\times$ too small). The correct treatment is **ETD (exact Ornstein–Uhlenbeck integration per k-mode)**:

$$\hat{\psi}' = a_k\hat{\psi} + (1-a_k)\frac{E\hat{c}}{k^2} + i\,\sigma\sqrt{\frac{1-a_k^2}{2\gamma_k\Delta t}}\,\hat{\xi}, \qquad a_k = e^{-\gamma_k\Delta t s_\psi},\;\; \gamma_k = k^2/E,$$

which reproduces the exact per-mode stationary variance of the linear part for ANY Δt; the $c[\psi]$ screening stays explicit (per-step rate $\leq \sum_i z_i^2\bar{\phi}_i\,\Delta t$). The $k=0$ mode is gauged to zero every step. - The EXCHANGE field's explicit mass is $\sim 1/\chi N$, so small-χ systems (e.g. the pure-salt validation) need $\Delta t < \chi N$; violating this diverges regardless of the charge sector. - The explicit Hamiltonian term must be evaluated ANALYTICALLY off the saddle: $+(1/2EV)\int\psi\nabla^2\psi$ (equal to $-\frac{1}{2}\langle c\psi\rangle$ only when Poisson holds).

At $E=0$ the ψ stiffness is infinite: ψ stays pinned at zero.

L-FTS prototype (mean-field ψ): scheme per saddle iteration — after each propagator solve, (i) build $\bar{\phi}_i$, c (k-space $\hat{h}_i$ multiplications), (ii) update ψ toward the Poisson solution (below), (iii) take the usual Anderson-mixing step on the imaginary SPT fields. Both residuals enter the stopping criterion. ψ stays OUT of `aux_fields_imag_idx` and the AM compressor state.

Stabilized ψ update (important). Replacing ψ by the bare solution $E\hat{c}/k^2$ every iteration is an undamped fixed-point sweep whose linearized gain is

$$g(k) \;=\; \frac{E}{k^2}\sum_i z_i^2\,\bar{\phi}_i\,\hat{h}_i^2(k),$$

which exceeds 1 at long wavelengths for realistic E (e.g. $E=25$, $k_{\text{min}}=2\pi/L$) — the joint saddle loop then diverges. Use instead the preconditioned Newton update with the ideal (local) screening estimate $S_{\text{scr}}(k)=\sum_i z_i^2\bar{\phi}_i\hat{h}_i^2(k)$:

$$\hat{\psi} \;\leftarrow\; \hat{\psi} \;+\; \frac{E\,\hat{c} - k^2\,\hat{\psi}}{k^2 + E\,S_{\text{scr}}(k)}, \qquad \hat{\psi}(0)=0.$$

The fixed point is unchanged ($-\nabla^2\psi = Ec$, monitored via the residual $-\nabla^2\psi/E - c$), but the screening term in the denominator damps exactly the modes the bare sweep amplifies; for ideal 1-segment ions the update is a true Newton step. S_{scr} only sets the convergence rate, so the segment-fraction estimate suffices for polymeric charges.

5.6 6. Discretization notes

- All new operations are diagonal in k-space: $\hat{h}_i(k)$ multiplications and the E/k^2 Poisson solve. Use the existing FFT objects and the deformation-vector $|\mathbf{k}|^2$ tables (`Pseudo`) — oblique cells work automatically.
- **No real-space $\nabla\psi$ is ever needed.** The electrostatic energy is evaluated by Parseval, and

at the saddle the substitution $\hat{\psi} = E\hat{c}/k^2$ eliminates the gradient entirely:

$$\frac{1}{2E}\int \nabla\psi\cdot\nabla\psi\, d\mathbf{x} = \frac{E}{2}\,\frac{V}{M^2}\sum_{\mathbf{k}\neq 0} w_t\,\frac{|\hat{c}_{\mathbf{k}}|^2}{k^2} = \frac{1}{2}\int c\,\psi\, d\mathbf{x}$$

(unnormalized `rfftn`; w_t = 2 interior / 1 edge modes, as in `wtmd.py`). Only $h_i * \psi$ requires an inverse transform. A real-space gradient (component-wise $ik_\alpha\hat{\psi}$) first becomes necessary for the deferred Maxwell-stress term. The identity holds at a Poisson saddle, where ψ is real; the term entering the recorded H is its negative per volume, $-\frac{1}{2V}\int c\,\psi$ (see §4). Off the saddle the CL code evaluates the analytic expression `0.5*mean(psi*lap(psi)/E)` instead.

- Smearing widths must satisfy $a_i \gtrsim dx/2$ **per species** — the binding constraint is $\min_i a_i$; below that the regularization becomes grid-dependent (same issue class as the bond-function study). With finite ζN this applies to ALL species, not just charged ones.
- Per-species radii compose automatically: the effective i-j pair kernel is $\hat{h}_i(k)\hat{h}_j(k) = e^{-(a_i^2+a_j^2)k^2/2}$, i.e. an effective pair smearing width $\sqrt{a_i^2+a_j^2}$ — Gaussian widths add in quadrature, so mixed radii (small ions, fat polymer segments) need no special treatment. Consequence for validation: with unequal radii the RPA/Debye–Hückel comparison must keep the per-species $\hat{h}_i^2$ factors separate (they only factor out as a common $\hat{h}^2$ when all radii are equal); the $k \to 0$ Debye limit is radius-independent ($\hat{h}_i(0) = 1$), which is why it is the safe unit test.
- Charge–charge structure factor $S_{cc}(k)$ from $\hat{c}(k)$: record alongside the existing $S(k)$.
- Stress with electrostatics (Maxwell term from ∂_L of the explicit electrostatic action $-\int \nabla\psi\cdot\nabla\psi/2E$) and with the $\hat{h}_i(k)$ cell dependence: defer; raise on `box_is_altering=True`.

5.7 7. Parameter dictionary (target)

```
"zeta_n": 100.0,                    # compressible Helfand penalty (existing param)
"charges":  {"P": 1.0, "S": None, "C": -1.0, "SP": +1, "SM": -1.0},
"radiuses": {"P": 0.025, "S": 0.025, "C": 0.025, "SP": 0.025, "SM": 0.025},
"bjerrum_e": 10000.0,               # E = 4 pi l_B rho_0 N R0^2   (NEW)
"chi_n": {"P,S": 50},               # full-matrix SPT; ion rows zero
# NO "chi_monomers" needed; NO incompressibility constraint
# molecules: counter-ion fraction computed from electroneutrality, not free
```

`None` charge = neutral species (no ψ coupling). `radiuses` now apply to ALL interactions (χ, ζ, Coulomb) of that species — in the compressible fluctuating model neutral species need smearing too.

The parameter dictionary above describes the general bulk formulation. The finite canonical and grafted fixtures use the explicit parameters in their protocol sections; these models are not silently substituted for one another.

6 Implementation checks and same-model MF/CL benchmark

6.1 1. Implemented changes

- **Partition checks:** eight CPU/CUDA, continuous/discrete, normal/reduce-memory implementations replace absolute errors of magnitudes with **complex relative errors**. Contour Q values are compared with the Q actually used for density normalization. The existing 1e-7 tolerance is unchanged. Zero/nonfinite Q also fails.
- **Strong-field reference:** models/stable_grafted_chain.py adds NumPy/CuPy Gaussian bond matrices. Chain propagation stores magnitudes and logarithmic scales separately; densities use one partition function. Complex propagators and densities are not clipped. Hamiltonians

use `log Q` to avoid underflow of Q itself. Only rounding errors in the positive real Gaussian kernel are removed during kernel construction.

- **Quenched grafting:** each lateral anchor is normalized by its own Q and uses `mean(log Q_anchor)`, distinct from annealed grafting with a uniform source and global Q. Zero width fixes the anchor to one grid cell; finite width averages its z location over the source.
- **CL diagnostics/restart:** H and structure functions use consistent pre-update fields, densities, and Q; checkpoint densities correspond to stored post-update fields. Separate psi RNG state, graft/model settings, shifts, and log Q are saved. `initial_params_json` preserves integer graft keys. At `E=0`, psi and electric energy are fixed to zero. ETD coefficients use accurate `expm1` at small arguments.
- **PRL MF comparison:** removed the invalid `psi -= psi[-1]` shift at fixed salt fugacity. PB nonconvergence fails; `solve()` reports final self-energy and Poisson residuals as well as W. Cell-centered integrals consistently use `sum * dz`. Direct chain calculations are selectable. Existing `height()` is explicitly a **2 <z> moment estimator**.
- Added an assertion where the Debye–Hückel test previously printed failure but exited successfully.
- **Mixed-valence Γ:** the external production driver `dh_salt_runs/test_brush_cl.py` now multiplies C and SP excess densities by `z_counter` and `zplus`, respectively. Mean/block Γ and error bars use the same corrected expression; species block profiles and sample counts are now saved.

Existing FFT/DST propagation paths were restored to their state before automatic correction. Relative partition checks and the Python direct backend remain. The direct calculation is a validation alternative. Default brush settings remain `spectral`/`annealed` for compatibility. The new backend supports single-block linear homopolymers with one segment species; it does not optimize large 3D production performance or support branched/multiblock chains.

6.2 2. Deterministic validation results

Check	Result
Installed `ctest -L basic`	56/56 passed, 36.7 s
New `TestPartitionFunctionCheck`, complex nonperiodic mirror	2/2 passed
Existing CL neutral reduction	Final w difference 0
Existing Debye–Hückel checks	18 combinations passed
Continuous Gyroid SCFT	221 iterations, residual `8.04e-9`, H `-0.674220654`
Quenched CL force–H finite differences, real/imaginary field std=5 each	Maximum error `9.99e-11`
CPU/CUDA densities at identical fields	Maximum difference `5.62e-16`
Fixed-anchor column density	Maximum deviation from 1: `2.56e-16`
Annealed anchors at the same lateral field	Column CV `1.084`; the ensembles differ
4-step uninterrupted run versus 2-step checkpoint restart	Maximum field/density/psi difference `5.33e-15`; identical psi RNG state
Stored density versus recomputation from stored fields	Maximum difference `5.59e-16`
Initial H time-step consistency	Error 0
E=0, periodic/wall	Removes preexisting nonzero psi and returns finite H/densities
Fixed-fugacity Donnan check	Potential `-0.2161641104`, maximum charge error `1.39e-17`
Nonconvergence with fixed incorrect self-energy	Reports failure after the iteration cap

The strong-field counterexample uses N=50, nz=704, L=6, a **Dirichlet image** bond kernel with absorbing ends, and per-chain W=100 for z<3. Its original field, grid, and chain parameters were not weakened.

Calculation	Q / log Q	Mean height	Mass error
Existing spectral CPU	Q=`-1.797e-18`	4.183237	2.55%
Existing spectral CUDA	Q=`-1.599e-18`	4.201706	1.70%
New direct CPU/CUDA	log Q=`-62.9319184541`, Q=`+4.667e-28`	3.076019224	At most `5.6e-15`

Updated partition checks reject both erroneous spectral cases and accept zero-field controls. Adding a uniform `1000+20j` to per-bead potentials changes direct-backend densities by at most `5.4e-14`. This distinguishes **small Q** from a **spatially inaccurate propagator**.

6.3 3. Connecting MF and CL for the same model

studies/validation.py uses a canonical five-species model in 2D: 8×32, L=(1.2,2), N=10, E=25, $\zeta N = 20$, $\chi_{PS} N = 10$, with C=1 and SP=3 valences. MF and CL use identical new quenched grafting and truncated-bottom/reflecting-top bonds. Dynamic stabilization is disabled with `alpha_ds=0`. **These benchmark numbers are not the PRL physical conditions.**

The full active-field MF residual is `5.14e-12`; MF $\langle z \rangle = 0.4518632443$.

nbar	dt	seed	Production Langevin time	CL mean height ± estimated SE
1e4	.05	51	400	.451938 ± .001787
1e4	.025	51	400	.451050 ± .001566
1e4	.05	52	1900	.452344 ± .000832
1e4	.025	52	1900	.452869 ± .000809
1e8	.05	51	400	.451842 ± .000180
1e8	.025	51	400	.451745 ± .000153

SE is the larger of the 20-block estimate and the initial-positive-autocorrelation-sum estimate. All six runs have mass errors `<3.34e-16` and contour partition errors `<2.56e-15`. Long-run imaginary means are statistically consistent with zero. The two dt values agree within current errors; increasing nbar suppresses fluctuations and approaches MF. This checks consistency under these conditions; **it does not establish dt→0 extrapolation or correct strong-coupling CL convergence**. Drift samples, complex block profiles for all species, and final checkpoints are preserved.

Refining the MF grid to 16×64 gives $\langle z \rangle = 0.4476186900$, a **0.94%** change. The fixed-cell anchor representation also changes with the grid; these are not continuum values. Besides the validation code, the external production driver's effective-charge calculation was corrected to use species valences. Mean Γ for 26 existing N=50 results was recalculated from original profiles and saved as separate copies. For example, `n5f_0p01_s12345.json` changes from $\Gamma = -0.07406685$ to $+0.23989397$. Historical mixed-valence results lack species block profiles, so corrected error bars cannot be reconstructed. Recalculated copies mark `gamma_blocks` and `gamma_err` as `null` and retain old values in metadata. Original files are unchanged. Correcting valences does not validate historical trajectories or agreement with PRL observable definitions. See the regression/recalculation summary and recalculated results.

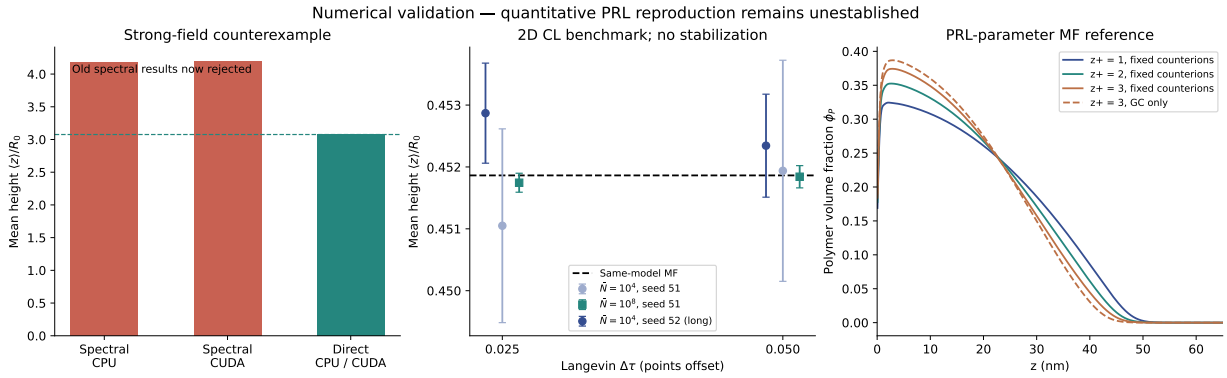


Figure 1: Same-model mean-field/CL bridge and deterministic validation. This does not certify general CL equilibrium sampling.

7 Exact finite-action benchmark for native CL

Follow-up: Reduced CL diagnosis extends this campaign with 21 trajectories and direct field quadrature. Smaller steps, longer sampling and alternative/adaptive integrators retain the discrepancy. Longer E=0 controls also resolve a smaller bias that the initial campaign below could not distinguish. The historical results below are preserved.

Positive-marginal and stationarity checks subsequently recovers the same exact averages with a positive real marginal obtained by analytic integration of repulsive fields. It also shows that simple stationarity checks and removal of near-zero observations cannot certify or repair the original biased CL data.

This benchmark tests the existing native discrete CL path against a positive particle partition sum. No FFT implementation, production drift, or existing test parameter is changed. It does not reproduce the PRL model or establish a continuum limit.

7.1 Model and independent reference

The unit-length, reflecting 1D box has two or four equally weighted cells. It contains one four-bead P chain, two monovalent C particles, one trivalent SP, one monovalent SM anion, and one neutral S particle. P has valence −1 per bead. The first P bead is fixed at cell 0; its other three beads are free. Thus the system has nine beads and zero total charge. There is only one graft anchor; the annealed/quenched lateral distinction does not arise in this 1D fixture.

With $N_{\text{ref}} = 4$, $V = 1$ and $C = \sqrt{\bar{N}} = 9/4$, the canonical particle multiplicities CVf_p/α_p are exactly 1, 2, 1, 1, 1. The volume fractions are 4/9, 2/9, 1/9, 1/9, 1/9; $\chi_{PS}N = 2$, $\zeta N = 4$, and all smearing radii are 0.1. At this coarse spacing the smearing is not continuum-resolved; both reference and CL use the same finite cosine multipliers.

Let $n_t(j)$ be the bead occupancy of species t in cell j. The microscopic volume fraction is $\phi_t(j) = n_t(j)/(CN_{\text{ref}}\Delta v)$, with $\Delta v = 1/M$. Using the smeared densities $\bar{\phi}_t$, the particle interaction energy is

$$U = CV\Big[\chi_{PS}N\,\text{mean}(\bar{\phi}_P\bar{\phi}_S) + \frac{\zeta N}{2}\,\text{mean}\left(\Big(\sum_t \bar{\phi}_t - 1\Big)^2\right) + \frac{E}{2}\,\text{mean}(c\,Gc)\Big],$$

$$c = \sum_t z_t\bar{\phi}_t, \qquad G = (-\nabla^2)^{-1} \text{ on the nonzero modes.}$$

Here mean averages over the M cells. All self terms implied by this finite field action are included. No separate self-energy subtraction, local incompressibility, or salt reservoir is added. The bond multiplier is $e^{-k^2\,ds/6}$, the smearing multiplier $e^{-a^2k^2/2}$, and the multiplier of G is $1/k^2$ except at $k = 0$, where it is zero. The reference constructs these matrices explicitly with cosine modes rather than calling the production transforms.

Each configuration has the positive weight $e^{-U}\prod_b B_b$, where B_b are the weights of its three bonds. Enumerating labelled C particles supplies their correct occupancy multiplicity; the canonical $2!$ is a configuration-independent factor and cancels from averages. The 2-cell reference sums all $2^8 = 256$ configurations; the 4-cell reference sums all $4^8 = 65{,}536$. Profiles are therefore obtained without Monte Carlo uncertainty.

Two preliminary identities check the mapping:

- At complex fields with real and imaginary standard deviation 5, enumerating noninteracting chains/particles with those external fields agrees with the native conditional densities: maximum error 3.09e-14 (4 cells, CPU) and 7.74e-16 (2 cells, CUDA). Native partition consistency also passes.
- Eliminating the quadratic auxiliary and electrostatic fields reproduces U above: maximum per-CV error 4.45e-15 in the checked configurations.

7.2 Stochastic comparison

The initial campaign has 22 parameter-specific SLURM jobs:

- 2 cells, E=0/25/100, dt=.02/.01/.005, stabilization=0, seeds 71/72.
- 2 cells, E=25, dt=.01, stabilization=.05, seeds 71/72.
- 4 cells, E=25, dt=.01, stabilization=0, seeds 71/72.

Each trajectory lasts 2000 Langevin units, discarding the first 200. Profiles, bare and stabilized drift maxima, and partition cancellation ratios are saved every 0.1 units. Native partition checks and species mass checks remain active. Failures preserve the offending auxiliary fields and psi; a finite trajectory or a successful process exit alone is not treated as a correctness test.

Two follow-up jobs use 2 cells, E=25/100, dt=.0025, stabilization=0 and seed 173, lasting 1000 units and discarding 100. These extend the time-step check; their shorter physical duration must be considered when comparing error bars. The original seeds 71/72 are reported separately: the existing psi generator uses seed+1, so these trajectories share one generator seed across different field components. They are not pooled as independent replicates. Seed 173's streams do not overlap those seed assignments. Production RNG code is unchanged.

The cancellation diagnostic is $|Q(W)|/Q(\mathrm{Re}\,W)$ using the same finite bond operator. It is invariant under uniform real field shifts, unlike the raw $|Q|$. A small ratio indicates cancellation near possible drift poles; a finite-sample histogram alone does not establish a CL correctness criterion.

For each observable the analysis reports 10/20/40-block errors, autocorrelation times, estimated effective sample counts, and imaginary means. The reported SE is the largest of those block estimates and the autocorrelation estimate. These are empirical uncertainty estimates, not rigorous bounds. Height and all species cell densities are compared with the exact reference. Distinct cells and species have correlated errors, so the largest standardized deviation is a diagnostic rather than an independent multiple-testing confidence level.

7.3 Results

All 24 trajectories completed, including their native partition and species mass checks. Maximum species mass error was 3.54e-13. The six E=0 controls agree with the reference: every checked real observable differs by at most 1.51 estimated SE. Charged trajectories do not show comparable agreement.

For the two-cell fixture at dt=.005 with stabilization off, the monovalent counterion volume fraction in cell 0 is:

E	Exact	CL seed 71 ± SE	CL seed 72 ± SE
0	.19243640	.19278850 ± .00033426	.19282432 ± .00035266
25	.20787987	.20933511 ± .00027945	.20926531 ± .00024669
100	.20718911	.21294327 ± .00058985	.21359478 ± .00081007

These E=25/100 discrepancies are 5.21–9.76 estimated SE. Their counterion autocorrelation estimates correspond to about 337–594 effective samples per trajectory, rather than treating all 18,000 stored values as independent. At E=100 the mean heights are .408940 ± .002489 and .409227 ± .002591, versus exact .39815593. E=25 seed 71 instead has a large height error estimate (.409159 ± .009108), illustrating the sensitivity of observables to rare events. The dt=.01 runs with stabilization=.05 also retain counterion discrepancies; that tested stabilization value does not resolve the comparison.

The finer-step follow-ups (dt=.0025, seed 173, stabilization off) give:

E	Observable	Exact	CL ± SE	Difference / SE
25	C density, cell 0	.20787987	.20926778 ± .00057137	2.43
25	SP density, cell 0	.12889864	.12650501 ± .00078950	3.03
100	C density, cell 0	.20718911	.21308439 ± .00112541	5.24
100	Mean height	.39815593	.40964106 ± .00355063	3.23

The E=100 counterion discrepancy persists at the smallest tested time step with disjoint seed assignments. Its central value does not move toward the exact result over dt=.02 to .0025 within the observed variation. These data still do not provide a controlled dt→0 extrapolation. In particular, the two follow-ups have half the production time of the original trajectories, and rare large density excursions make empirical error estimates uncertain.

The four-cell fixture is a separate spatial discretization, with its own exact reference. Its height agrees within the estimated uncertainty, but P density in cell 1 does not: exact .47375505, versus .45115654 ± .00333992 and .44947711 ± .00337511 at dt=.01. A height-only acceptance check would miss this.

At all 24 final checkpoints, independent enumeration in the saved complex fields reproduces the saved native densities to at most 9.67e-15. This checks the chain/particle density calculation at those states, not every intermediate state or the rarest drift events. It provides no reason to change native FFT code to resolve these stochastic discrepancies.

Charged trajectories also sample stronger drift excursions and smaller partition cancellation ratios than the E=0 controls. For example, E=25, dt=.005, seed 71 reaches a sampled bare drift of 455.45 and a cancellation ratio of 6.96e-4. These observations motivate investigation of rare events and drift poles; they do not establish the cause or prove a tail asymptotic law.

The charged CL comparison is not validated at the tested settings. Finite time steps and empirical errors in distributions with rare large events do not prove an incorrect dt→0 CL limit. Conversely, finite trajectories, conserved mass, small imaginary averages and consistent chain partitions do not justify accepting these averages or claiming PRL reproduction. The next question is to distinguish integration bias, insufficient exploration of rare events, and possible incorrect CL convergence for this finite model before using target PRL parameters.

7.4 Reproduction and evidence

studies/exact.py implements the reference and the run; studies/cl_analysis.py analyzes saved samples. studies/cl_diagnostics.py independently checks conditional densities at final saved fields.

```
python devel/charged_polymers/workflow.py legacy validate_clfts_exact.py \
  --out /shared/check --sites 4 --deterministic-only
python devel/charged_polymers/workflow.py legacy analyze_clfts_exact.py \
  /home/yongdd/polymer/dh_salt_runs/exact_cl_20260907
python devel/charged_polymers/workflow.py legacy check_clfts_exact_checkpoint.py \
  /home/yongdd/polymer/dh_salt_runs/exact_cl_20260907
```

Long runs must use the per-configuration `run.sh` SLURM scripts in `/home/yongdd/polymer/dh_salt_runs/exact_cl_20260907`. That directory preserves the runner snapshot, source hashes, parameters, checkpoints, raw samples and job identifiers. Initial submission failed during conda shell activation before simulations began; `run.sh` now enables nounset after conda activation, and those environment failures are separately recorded. Simulation jobs 333502–333525 all finished COMPLETED with exit code 0:0; this is execution status, not a successful CL correctness comparison. The numerical results are in `analysis.json`, conditional-density comparisons in `checkpoint_checks.json`, and plots in `exact_cl_validation.png/.pdf`. `source_unchanged_check.json` confirms that all ten recorded production/runner source hashes match the launch versions, including the native FFT solvers.

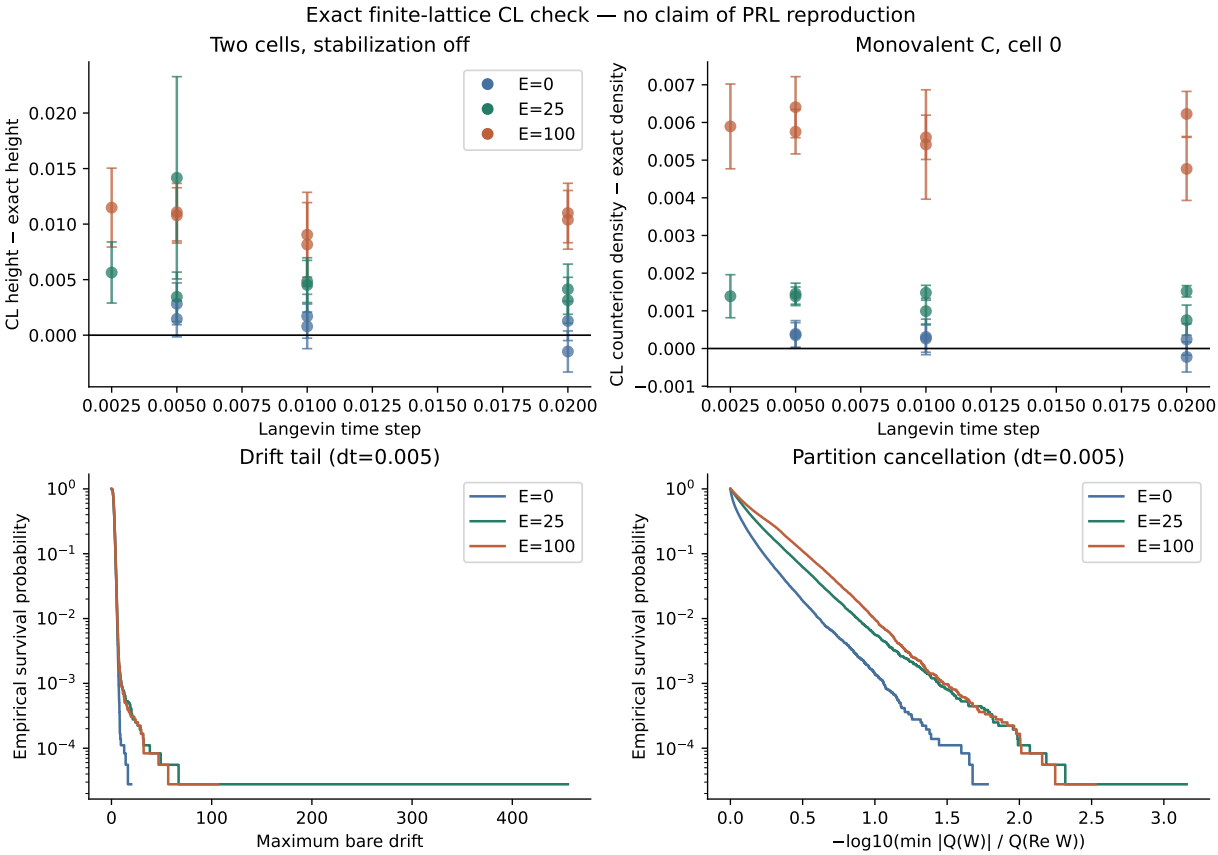


Figure 2: Native CL compared with the exact finite-particle sum. Later, longer controls resolve additional small biases.

8 Reduced CL: time-step, integrator and pole diagnosis

Further investigation: Positive-marginal and stationarity checks adds an exactly equivalent positive marginal sampler. Twelve real-Langevin runs recover the exact averages. Six original CL trajectories pass the tested low-order stationarity identities despite biased means; excluding near-zero observations does not remove that bias. The original CL path remains unchanged.

The current CL dynamics fails the exact finite-model sampling benchmark. The discrepancy persists with much smaller steps, longer trajectories, a different integrator, adaptive steps and independent seeds. The action, conditional densities, projected noise and original one-step update pass independent checks. Actual partition zeros occur within the sampled field region and generate the observed drift extremes. These results strongly support incorrect CL convergence in this fixture, rather than an FFT error or an incorrect particle reference. They are numerical evidence, not a formal proof of the infinite-time, zero-step limit or a complete boundary-term analysis.

No production FFT, CL drift or existing calibrated test parameters were changed. The new implementations are diagnostic Python scripts only. This is the same two-cell canonical model described in Exact finite-action benchmark, not the PRL model and not a continuum calculation.

8.1 Exactly reduced dynamics

Write each field as its spatial mean plus $x\,(1,-1)$. In this canonical model, uniform species fields multiply single-particle partitions by constants that cancel from normalized densities. Uniform auxiliary modes have decoupled Gaussian dynamics; only their differences affect the spatial observables. Eliminating those uniform modes therefore leaves the same marginal stochastic dynamics, without changing interactions, charges, smearing or the bond matrix.

There are four remaining complex amplitudes: native active auxiliary indices 0, 3, 4, followed by

ψ. These are called `field_0` through `field_3` in the saved analysis. In particular, `field_2` is the amplitude of native auxiliary mode 4, not the frozen native mode 2.

The reduced runner explicitly sums the eight conformations of the four-bead grafted chain, grouped by the occupancy of cell 0. Point-particle densities use their exact two-cell tanh expression. The noise variance of a half-difference is the native per-cell variance divided by two, namely $2\,dt/C$ before the mobility factor. The imaginary noise directions and mobilities are preserved.

Check	Maximum discrepancy
Native versus reduced conditional densities; 64 complex fields, real/imaginary std=5	4.81e-14
Native versus reduced drift at those fields	7.36e-14
Partition cancellation ratio	1.00e-15
Projected noise variance	0
Actual `BrushChargedCLFTS.run` step with the same supplied noise increments	4.97e-15

The native tests call `check_total_partition()` and compare all species. The one-step comparison reads the native post-update checkpoint, including ψ. Additional checks cover E=0 and ψ mobility 0.1.

8.2 A second independent reference: the original field integral

Gauss-Hermite quadrature directly integrates the original field contour after the uniform modes have been integrated out. Its integrand is $Q_P\,Q_C^2\,Q_{SP}\,Q_{SM}\,Q_S$ times the Gaussian field factors. The implementation forms density numerators without dividing by Q and does not evaluate $\log Q$. It is an entire integrand, including at partition zeros.

Besides densities, it integrates each field amplitude and its square. Their particle references follow by completing the Gaussian square for every one of the 256 particle configurations. Let $\delta\phi_t$ be the microscopic half-difference of species t between the two cells, $s = e^{-a^2\pi^2/2}$ the smearing multiplier of that mode, A_{ti} the auxiliary-field matrix (`matrix_a`) and h_i the quadratic coefficient of field i (`h_coef_mu2[i]`). The conditional Gaussian centers are

$$\mu_i = -\frac{s}{2h_i}\sum_t A_{ti}\,\delta\phi_t, \qquad \mu_\psi = \frac{E}{\pi^2}\,s\sum_t z_t\,\delta\phi_t.$$

The corresponding analytic covariances of the stored complex fields are $1/(2Ch_i)$ and $-E/(C\pi^2)$. Averaging the centers and their squares plus these covariances over the positive particle weights gives the exact field moments. Thus the comparison also uses entire observables that do not themselves contain a partition denominator.

E	Hermite order per active dimension	Maximum error across densities and field moments
100	20	1.479e-3
100	32	2.684e-10
100	48	2.788e-15
0	32	2.443e-15

This independently confirms the particle-to-field mapping, including the finite-grid self terms and the canonical multiplicities.

8.3 Stochastic results

Twenty-one additional trajectories and four quadrature jobs completed:

- E=100: original LM/ETD and additive-noise Heun, dt=.0025/.000625, seeds 301/401, total time 10,000 with 1,000 discarded.
- E=0/25: both integrators at dt=.000625, seed 301, the same duration.
- E=100: Heun with ψ mobility .1/10 at dt=.000625, seed 301, the same duration.
- E=100: dt=.00015625, time 20,000 with 2,000 discarded; LM/ETD seed 501 and Heun seeds 501/601.
- E=0: dt=.000625, time 50,000 with 5,000 discarded; Heun seed 501 and LM/ETD seed 601.
- E=100: adaptive Euler at maximum dt=.000625/.00015625, seed 701, time 20,000 with 2,000 discarded. The step is chosen from the current drift before noise is drawn: $dt_{\text{actual}} \leq dt_{\text{max}} \min(1,\, 10/\max|\text{drift}|)$. Noise scales with $\sqrt{dt_{\text{actual}}}$. Observations are taken at fixed physical times, not weighted equally per adaptive step. No clipping, rejection or stabilizing force is added.

All observations are spaced by .1 Langevin time. Errors are the maximum of 10/20/40-block and autocorrelation estimates, as in the earlier benchmark. Runs that share a seed are paired comparisons, not independent replicates. Different fields use successive PCG64 draws within the reduced runner; cross-run seeds are separated. The reduced stochastic trajectories need not match the native trajectories point for point to have the same noise law.

At E=100, dt=.00015625, with ψ mobility 1:

Method, seed	C density in cell 0 ± SE	Auxiliary amplitude `field_2` ± SE
Exact	.20718911	.65699013
LM/ETD, 501	.21321148 ± .00020726	.56404950 ± .00185565
Heun, 501	.21314436 ± .00021742	.56421569 ± .00185824
Heun, 601	.21312338 ± .00020704	.56384330 ± .00188818
Adaptive Euler, 701	.21331872 ± .00019318	.56490673 ± .00190322

The C-density discrepancies are about 27–32 estimated SE; the entire auxiliary field mean differs by about 48–50 estimated SE. The various integrators agree with each other much better than they agree with the exact reference. The adaptive fine-step run reaches the full physical duration in 128,130,102 steps. Changing ψ mobility by factors of ten also does not restore agreement at the tested settings. These findings cannot be explained simply by the original LM/ETD time step or by statistical noise in a single short trajectory.

The longer E=0 controls also fail this stronger comparison. At dt=.000625, the exact C density .19243640 is compared with .19302000 ± .00006632 (Heun, seed 501) and .19307754 ± .00007713 (LM/ETD, seed 601), differences of 8.8 and 8.3 estimated SE. The earlier 2,000-time-unit controls did not resolve this smaller bias. Their reported measurements remain unchanged, but their apparent agreement must not be used as evidence that this finite E=0 CL model is valid. This does not invalidate the neutral-reduction test: that checks equality of two code paths, not correctness of their equilibrium distribution.

8.4 Actual partition zeros and drift poles

For the smeared polymer field half-difference h, the chain partition is

$$Q_P(h) = c_0 e^{h/2} + c_1 + c_2 e^{-h/2} + c_3 e^{-h},$$

where the positive coefficients c_n sum the eight bond-weighted conformations. Factoring out $e^{h/2}$ gives a cubic polynomial in $e^{-h/2}$. Its exact roots locate partition zeros at

$$h = 0.9614601145 \pm 2.7421589828\, i, \qquad h = 1.26847297 + 2\pi i,$$

and their repeats by $4\pi i$. Each point-particle partition also has zeros at $h = (2n+1)\, 2\pi i$. These are zeros of the mathematical partition functions, independent of FFT evaluation. They produce poles in derivatives of $\log Q$.

The strongest recorded drift states lie close to P or SP zeros. For example:

- E=0, Heun seed 301: polymer $h = 0.96143145 + 2.74221821\,i$, only 6.58e-5 from a Q_P zero; drift magnitude 10,167.
- E=100, fine adaptive Euler: polymer $h = 0.96145934 - 2.74216402\,i$, only 5.09e-6 from a Q_P zero; drift magnitude 131,319. Its time step is reduced accordingly.

At the maximum-drift state of each of the 21 trajectories, the native solver still passes its unchanged partition check. Comparing native densities with full conditional enumeration gives a maximum scaled error of 1.98e-10, using max $|\text{difference}|/(1 + \max|\text{reference density}|)$. Densities can be of order 1e5 near these poles, so absolute errors are larger there (up to 1.73e-5 in the fine adaptive example); they should not be described as uniformly 1e-14. The independent reduced calculation and the native solver agree despite their different evaluation methods, and both encounter the same true zeros.

The connection between sampled drift poles and possible incorrect CL convergence is established in the CL literature; merely having a pole does not prove failure in every model. See Aarts et al., JHEP 05 (2017) 044 and Nagata, Nishimura & Shimasaki, PRD 94 (2016) 114515. Here the independent exact averages establish a sampling discrepancy, while the pole analysis identifies a concrete mechanism consistent with it. A full boundary-term calculation or mathematical classification of invariant measures has not been performed. Finite-sample error bars and tail plots alone are not such a proof.

8.5 Consequences and remaining work

The deterministic implementation checks support the existing action-to-drift and chain computations in this fixture. They do not certify that its CL stationary distribution represents the intended field integral. The tested changes in integrator, step size and mobility do not supply a correction. No production patch is justified by tuning the output to the reference.

Before accepting a production CL result, a sampling formulation or a clearly bounded parameter regime must pass independent equilibrium checks. For this finite fixture, direct quadrature already supplies a correct field-integral reference; it is not a scalable replacement for a 3D production sampler. This benchmark does not prove that every CL parameter regime fails, and it does not establish behavior at the physical PRL parameters. The separate PRL model/ensemble/boundary mismatches in Implementation checks also remain. PRL reproduction is not complete.

8.6 Code and evidence

- studies/reduced.py: reduction, native comparisons, fixed and adaptive integrators, exact field moments.
- studies/cl_diagnostics.py: quadrature.
- studies/cl_diagnostics.py: analytic poles and native checks at drift extremes.
- studies/cl_analysis.py: correlated errors and plots.

Raw trajectories, parameters, source snapshots, RNG states, maximum-drift states, SLURM scripts and job statuses are under `/home/yongdd/polymer/dh_salt_runs/exact_cl_diagnosis_20260907`. The original 19 trajectories use snapshot v1; the two adaptive trajectories use v2. A subsequent deterministic check adds the explicit noise-variance assertion without changing either integrator. `production_unchanged_check.json` confirms the production source hashes are unchanged from the campaign launch.

Numba 0.67.0 and llvmlite 0.49.0 were installed only into that evidence directory's `python_deps`, not into the shared conda environment. Example:

```
conda activate polymerfts
export OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 OMP_MAX_ACTIVE_LEVELS=0 OMP_STACKSIZE=1G
ulimit -s unlimited
export PYTHONPATH=/home/yongdd/polymer/dh_salt_runs/exact_cl_diagnosis_20260907/python_deps
python devel/charged_polymers/workflow.py legacy diagnose_clfts_reduced.py --out /shared/check
↪  --checks-only
python devel/charged_polymers/workflow.py legacy analyze_clfts_diagnosis.py \
  /home/yongdd/polymer/dh_salt_runs/exact_cl_diagnosis_20260907
```

Use the saved per-configuration `run.sh` scripts with SLURM for long runs. All 25 campaign jobs

completed with exit 0:0; this describes execution status, not successful CL equilibrium validation. Plots are `integrator_diagnosis.png` and `polymer_partition_poles.png`, with PDF copies.

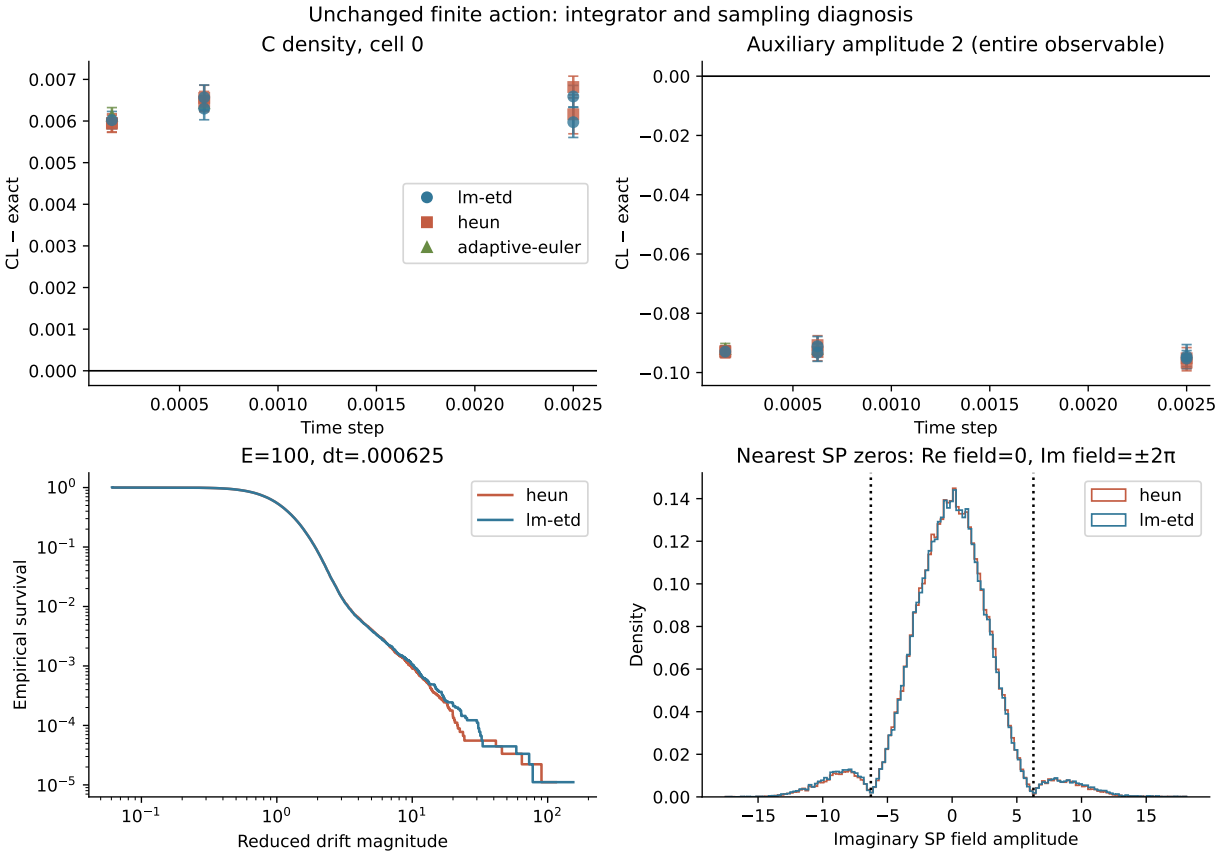


Figure 3: Reduced CL integrator and time-step comparisons against exact means.

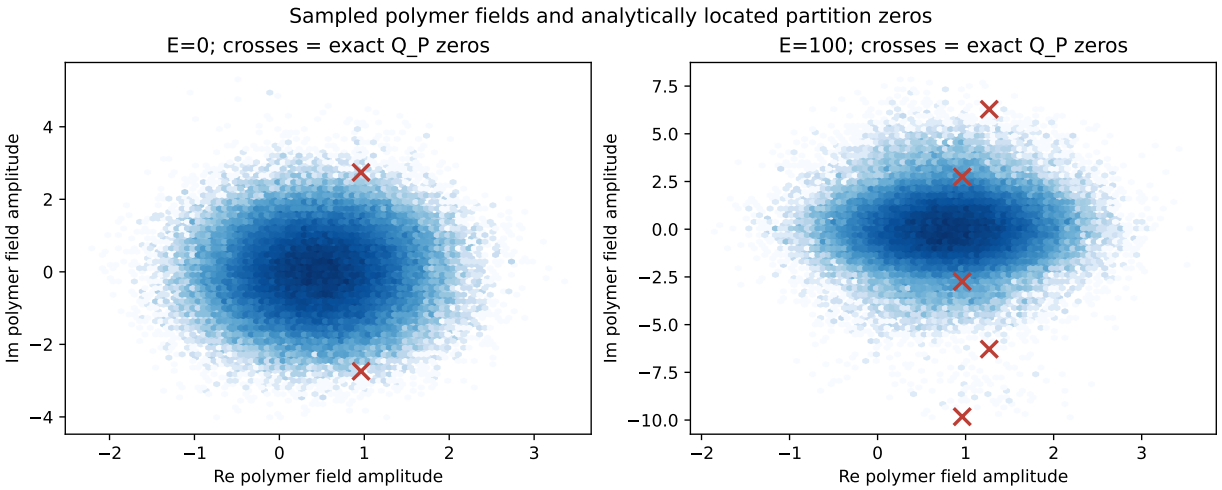


Figure 4: Exact polymer partition zeros and sampled field regions; poles alone are not a proof of incorrect convergence.

9 Positive marginal and stationarity diagnostics for the same action

The following Gibbs-sampling chapter extends this control to 4/16/32 cells without exhaustive marginal construction. PRL-specific calculations are in the separate PRL report.

Exact Gaussian marginalization restores agreement with the finite-model reference; the original CL trajectories remain biased. The new sampler changes the field representation while preserving the physical action. Twelve real-Langevin runs at E=0/25/100 agree with all 19 reference observables within 1.36 estimated standard errors. Six saved CL trajectories also show that their bias is not a direct averaging artifact from a few samples near partition zeros.

This extends Reduced CL diagnosis. It is a two-cell diagnostic, not a production CL repair or PRL reproduction. Production FFT, CL drift, physical parameters and existing calibrated tests are unchanged.

9.1 A positive marginal of the identical action

Use the four complex half-difference amplitudes of the earlier reduction: three active auxiliary fields (native indices 0, 3, 4) and ψ. For each of the 256 particle configurations, their action is quadratic plus linear in the fields. Integrate the two repulsive auxiliary amplitudes and ψ analytically. Retain only the attractive amplitude x (`field_0`) on its real contour.

Let a_i be the quadratic coefficients and, with s, A_{ti} and $\delta\phi_t$ as in the field-integral reference, $l_i = s\sum_t A_{ti}\,\delta\phi_t$, $q = s\sum_t z_t\,\delta\phi_t$ and $C = 9/4$. The integrated configuration weight apart from the retained field is

$$B_{\text{config}} = B_{\text{bond}} \exp\left[C\sum_{i=1,2}\frac{l_i^2}{4a_i} - \frac{CEq^2}{2\pi^2}\right],$$
$$p(x) \propto e^{-Ca_0x^2}\sum_{\text{config}} B_{\text{config}} e^{-Cl_0x}.$$

Here $a_0 = 1.25$ and $a_1, a_2 < 0$. Grouping configurations with the same integer occupancy difference $n_P(0) - n_S(0)$ of cell 0 reduces the sum to five terms. All coefficients are positive. Thus $p(x)$ **has no zero for real** x, and its logarithmic derivative is finite along the sampled path. This does not assert that its analytic continuation has no complex zeros.

The real Langevin drift is $-2a_0x - \langle l_0\rangle_x$ with noise variance $2\,dt/C$. The sampler uses additive-noise Heun. Densities and the eliminated field moments are conditional averages at x; x and x^2 are observed directly. Coefficients come from the bond weights and interaction parameters, without fitting to the exact means. No saddle approximation, interaction removal, clipping or parameter adjustment is used.

The last real Gaussian integral reproduces the original normalized particle weights to `5.56e-17`; numerical real quadrature reproduces all 19 observables to `2.56e-15`. A finite-difference check of the marginal log-density derivative at 64 random fields of std=5 has maximum error `1.58e-10`.

9.2 Sampling comparison

The 12 runs cover E=0,25,100; dt=.005,.00125; and seeds 901,1001. Each runs to Langevin time 20,000, discards 2,000 and records 180,000 observations spaced by .1. Error bars use the maximum of 10/20/40-block and autocorrelation estimates, as in the original benchmark. Seeds distinguish two replicates per setting; the same seeds across parameters are paired comparisons, not independent runs. The 19 observables are also correlated, not 19 independent statistical tests.

Every run completed, conserved each species' mass to `2.23e-16`, and agreed with the reference within 1.353 estimated SE for every measured observable: height, ten species/cell densities, four field means and four field squares.

At E=100, total time 20,000 with 2,000 discarded:

Sampling method, seed	dt or maximum dt	C density, cell 0 ± SE	field_2 mean ± SE
Exact particle sum	—	.20718911	.65699013
Original CL LM/ETD, 501	.00015625	.21321148 ± .00020726	.56404950 ± .00185565
CL Heun, 501	.00015625	.21314436 ± .00021742	.56421569 ± .00185824
CL adaptive Euler, 701	.00015625	.21331872 ± .00019318	.56490673 ± .00190322
Positive real marginal, 1001	.00125	.20719224 ± .00005738	.65695408 ± .00218514

field_2 means native auxiliary mode 4 in this reduced numbering. It is an entire field observable, so its original CL discrepancy does not arise from dividing the observable itself by a small Q.

The new real sampler also resolves the smaller E=0 discrepancy: at dt=.00125, seed 1001, C density is .19244016 ± .00007421, against exact .19243640. Original long CL runs gave approximately .19302–.19308 with SE .00007.

Comparison figure (PNG) and PDF show the original and new E=100 results. All 12 runs and 19 observables are in positive_comparison.csv in the evidence directory.

9.3 The existing CL bias survives exclusion of near-zero observations

For six saved CL trajectories, measure each configuration's minimum distance to any exact P or point-particle partition zero in its smeared field amplitude h, including the imaginary-period repeats. Analyze observations saved at fixed physical times. These distances are not the minima over every integration step.

For an observable O and the distance d of a sample to the nearest zero, decompose its measured bias as

$$\begin{aligned}\text{mean}(O - O_{\text{exact}}) = {} & \text{mean}\left[(O - O_{\text{exact}})\,\mathbf{1}(d \le \epsilon)\right] \\ & + \text{mean}\left[(O - O_{\text{exact}})\,\mathbf{1}(d > \epsilon)\right].\end{aligned}$$

Both terms divide by the original full sample count; they are not conditional means or a proposed corrected estimator. At $\epsilon = 0.1$:

CL run	Near-zero sample fraction	Total C-density bias	Near-zero contribution	Remaining contribution
E=100, Heun 501	.1789%	.00595525	.00000299	.00595226
E=100, LM/ETD 501	.1733%	.00602237	.00000265	.00601972
E=100, adaptive Euler 701	.2050%	.00612961	.00000096	.00612865
E=0, Heun 501	.0364%	.00058360	−.00000328	.00058689

The fourth E=100 run (Heun 601) gives the same conclusion. Over 99.94% of the E=100 C-density bias remains in the contribution from outside these small neighborhoods. The field_2 bias also remains almost entirely outside them. The result holds qualitatively at larger neighborhoods recorded in the data. This rules out simply removing a few extreme measurements as a remedy. It does **not** rule out the dynamical influence of brief pole encounters on the subsequent distribution, or prove which pole causes that distribution.

9.4 Low-order stationarity does not validate the sampled measure

Compute the unevolved holomorphic generators for $O = x_i,\ x_i^2$ with $i = 0, 2, 3$:

$$L\,x_i = f_i, \qquad L\,x_i^2 = 2x_i f_i + 2D_i.$$

Here f_i is the drift of x_i. For auxiliary amplitudes $D_i = \sigma_i m_i / C$, with the mobility m_i and the sign σ_i (+1 for a real, −1 for an imaginary field); for ψ it is $-m_\psi / C$, or zero at E=0. Negative D_i here accounts for imaginary noise in the stored complex coordinates. In all six trajectories, every tested real and imaginary part of $\langle LO \rangle$ lies within 1.686 estimated SE of zero, despite their incorrect equilibrium averages. The LM comparison concerns the continuous generator at its finite time step, not an exact finite-step stationarity identity.

Cutoff bulk averages and cutoffs in the imaginary directions after noise-axis rotation are saved as diagnostics. They are **not a complete surface-flux or evolved-observable boundary-term calculation**. A finite cutoff signal alone does not prove a nonzero limiting boundary term; a small uncut value does not prove correct convergence.

This limitation is consistent with Salcedo & Seiler, J. Phys. A 52 (2019) 035201, which discusses incorrect CL limits that satisfy Schwinger-Dyson identities. Their contour-classification theorem is for one complex variable; it has not been established for this four-variable fixture. Seiler's pole-boundary analysis also shows, in its simple models, why unevolved late-time diagnostics can miss pole effects. Neither paper supplies a completed boundary-term proof for this model.

9.5 Interpretation and evidence

The exact particle sum, original contour quadrature and new positive marginal sampler now agree on the same finite action. The original CL paths still give different means. Together with the earlier native drift/noise/update checks, this further supports a problem with the sampled CL measure in this fixture. Poles remain a concrete candidate mechanism; this experiment changes several field coordinates at once and does not isolate one pole or prove all CL fails.

The positive marginal is computationally cheap here because only 256 particle configurations exist. Building the corresponding conditional sums in a large 3D polymer system would not be cheap. This diagnostic therefore supplies a working finite-model control, not a scalable replacement for production CL. The existing CL path and the PRL model/ensemble comparison remain unvalidated.

New code:

- studies/marginal_sampling.py: exact marginal, deterministic checks and real Langevin sampling.
- studies/cl_diagnostics.py: generators, analytic pole distances and decomposition of measured bias.
- studies/cl_analysis.py: aggregate checks, CSV and standalone figures.

Evidence is under `/home/yongdd/polymer/dh_salt_runs/exact_cl_pole_resolution_20260907`. SLURM jobs 333564–333575 (12 sampling runs) and 333584–333589 (six analyses) all completed successfully. The initial six analysis submissions failed on an unsupported Numba `np.r_` expression before producing results; their logs are preserved separately. Replacing it by an explicit array initialization and checking compiled/Python agreement fixed the diagnostic only. No trajectories or physical parameters were changed to resolve that compile failure.

Each saved `run.sh` records the command, setup path and environment. Numba and llvmlite remain confined to the earlier campaign's `python_deps`. `analysis.json`, source snapshots, RNG states, SLURM logs and production hash checks preserve the evidence needed to reproduce this comparison.

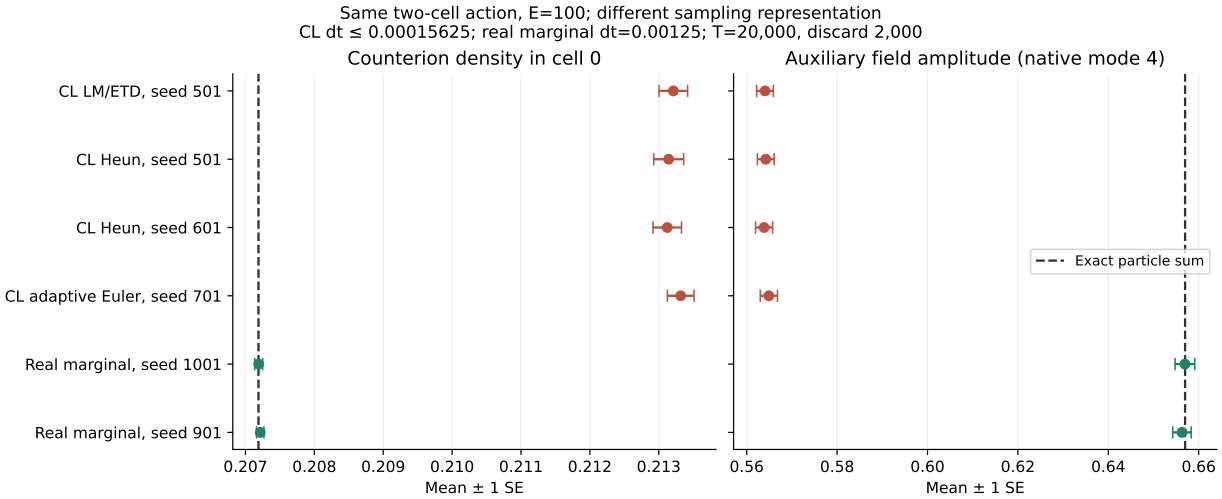


Figure 5: Same-action positive marginal recovers exact means while the original CL trajectories remain biased.

10 1D Gibbs sampling at 4, 16 and 32 cells

The earlier two-cell marginal retaining only a real field summed all particle configurations. Here the repulsive auxiliary field and ψ are integrated exactly, while **particle positions and the real field in the attractive channel are retained together**. This Gibbs sampler draws the real field from a Gaussian at fixed particles and updates each particle using its exact conditional probabilities at fixed field. It uses no finite-time-step Langevin integration. It is neither a field-only sampler with all particles integrated out nor the same dynamics as existing CL.

The equivalence of the finite action follows as below. Let x be the real field with its uniform mode removed, ℓ the corresponding attractive-channel combination of smeared densities, and $a > 0$ the positive field coefficient. Write the spatial mean as $\overline{f} = M^{-1}\sum_{i=1}^{M} f_i$ and the bond weight as $B(R)$:

$$U_{\text{rep}}(R) = U_{\text{particle}}(R) + \frac{C}{4a}\overline{\ell(R)^2},$$
$$P(R,x) \propto B(R)\,\exp\left[-U_{\text{rep}}(R) - C\overline{ax^2 + \ell(R)x}\right].$$

Integrating x cancels the added quadratic term exactly, recovering the original particle weight. The conditional field mean and covariance are

$$\mathbb{E}[x \mid R] = -\frac{\ell(R)}{2a}, \qquad \text{Cov}(x \mid R) = \frac{M}{2Ca}\left(I_M - \frac{11^{\mathsf{T}}}{M}\right).$$

The a in this section differs from the PRL Born radius. Particle updates include every interaction, self term, neighboring bead bond weight, and the fixed first bead. Each update sums M candidate positions, without enumerating all M^8 configurations.

A separate particle-only Gibbs control also integrates out the attractive field. Both samplers' energies were checked against direct evaluation of the original density action; particle conditional probabilities were independently computed for real fields with std=5. Existing FFT and CL drift were unchanged.

Conditions follow Exact finite-action benchmark: $N_{\text{ref}} = 4$, one 4-bead chain, two C(+1), one each of SP(+3), SM(-1), and S, $L = 1$, $\chi N = 2$, $\zeta N = 4$, smearing radius 0.1, and reflecting bonds. The first bead is fixed in the first cell.

- M=4/16/32 and E=0/100: real-field/particle seeds 1101,1201 and particle-only seed 2101, totaling 18 runs.

- Three additional M=4, E=25 runs directly match the historical native CL conditions.
- Each run: 2,000,000 sweeps, discard the first 200,000, save every 10 sweeps, yielding 180,000 samples. A sweep updates eight free particles and the conditional real field.
- Recorded quantities include height, species densities in every cell, local conditional field moments, and the mean and square of the sampled first cosine mode. Particle-only controls evaluate field moments analytically.
- SE is the larger of the existing 10/20/40-block and autocorrelation estimates. Species, positions, and moments are correlated; parameter comparisons sharing a seed are also not independent replicates.

Four-cell results were compared against the exact sum of 65,536 configurations. Every measured quantity across nine runs was within 2.781 estimated SE of its exact value. For 16 and 32 cells, the two samplers were compared without exhaustive enumeration; there is no independent exact-sum reference at these sizes. The largest difference between samplers across all runs is 3.049 combined SE. These descriptive statistics cover many correlated positions/observables; they do not mean that every individual 2-SE interval passed. The maximum energy identity error is 1.25e-14, conditional probability error 7.78e-16, and every sample conserves each species' mass.

P density in cell 1 at M=4, E=25:

Calculation	Mean ± estimated SE
Exact particle sum	.47375505
Real-field/particle Gibbs, seed 1101	.47113827 ± .00095049
Real-field/particle Gibbs, seed 1201	.47464444 ± .00083619
Particle-only Gibbs, seed 2101	.47479012 ± .00115897
Historical native CL dt=.01, seed 71	.45115654 ± .00333992
Historical native CL dt=.01, seed 72	.44947711 ± .00337511

Historical CL values were read from their original files without overwriting them. This compares the four-cell CL at its existing dt; the four-cell dt→0 limit was not newly investigated.

Mean height $\langle z \rangle / L$ at $E = 100$:

1D cells	Real-field/particle seed 1101	Real-field/particle seed 1201	Particle-only seed 2101
4	.360268 ± .000306	.360473 ± .000374	.360899 ± .000341
16	.332038 ± .000318	.331891 ± .000292	.332504 ± .000319
32	.327538 ± .000321	.328409 ± .000294	.328293 ± .000331

The exact four-cell height is .36049837. Changes from 16→32 cells exceed statistical errors, so continuum convergence is unverified. The spatial representation of the anchor fixed in the first cell also changes with the grid. This small system has eight free particles; it does not validate performance for an actual N=100 brush or the PRL model.

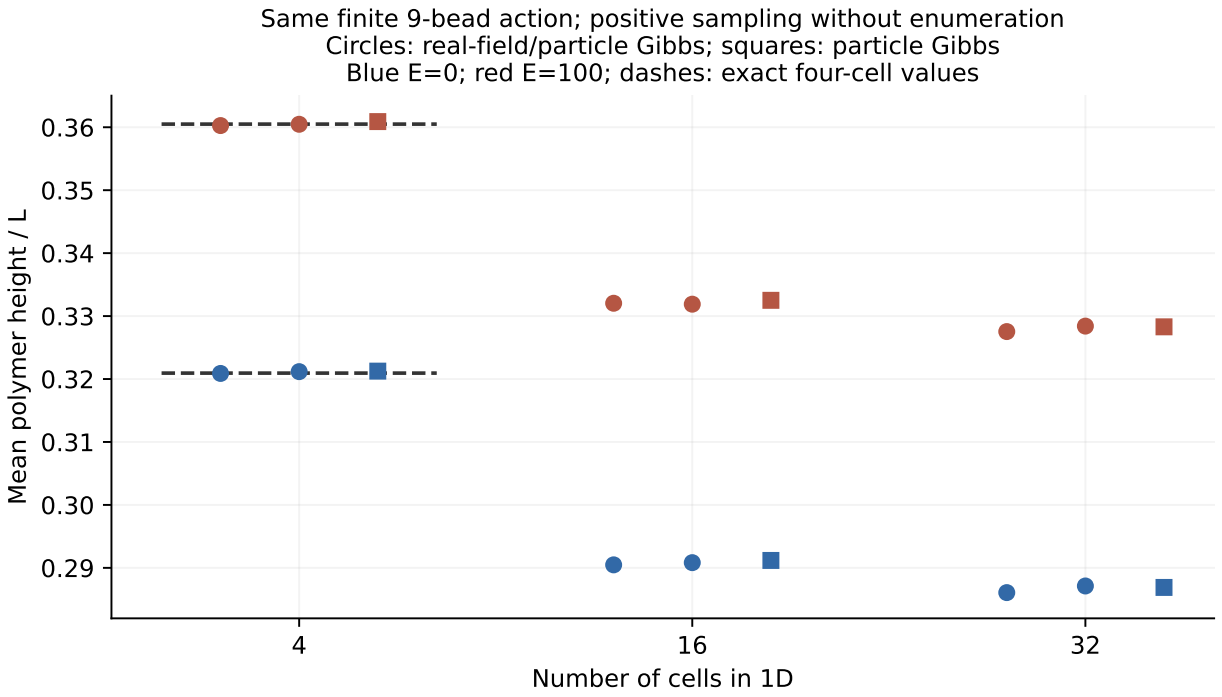


Figure 6: Small nine-bead Gibbs controls. This is separate from the N=100 planar brush campaign.

11 Drift-tail criterion and convergent regimes (2026-10-26)

Nagata, Nishimura and Shimasaki PRD 94 (2016) 114515 require the probability distribution of the drift magnitude to fall off exponentially or faster; a power-law tail lets boundary terms bias the averages. This chapter applies the criterion to the two-cell fixture of Exact finite-action benchmark with $\chi_{PS}N$, ζN and E as parameters. It answers two questions: whether the failing settings violate the criterion, and in which settings the CL averages agree with the exact sums. Production FFT, CL drift and existing calibrated tests are unchanged.

11.1 Method

The particles, the box, the smearing radius 0.1, $C = 9/4$ and $N_{\text{ref}} = 4$ are those of the benchmark; $\chi_{PS}N = 2$ in every run. The exact reference sums the 2^8 configurations with the energy U of the benchmark for the given $\chi_{PS}N$, ζN and E. The dynamics is that of the reduced diagnosis: four complex half-difference amplitudes, the three active auxiliary fields and ψ. studies/nagata.py evaluates it for independent walkers at once in NumPy. Before each run the reduced densities and drift are compared with the native solver at 32 complex fields with real and imaginary standard deviation 5; the largest difference over all runs is 3.5×10^{-14}.

Each run starts 2048 independent walkers at the saddle point, discards 20 Langevin time units and measures 100 units with additive-noise Heun at $dt = 0.000625$, that is 3.28×10^8 walker steps. The production LM/ETD integrator and $dt/2$ are checked in one convergent setting. Errors are the standard deviation of the walker means divided by $\sqrt{2048}$; the walkers are independent replicates. At every step the drift magnitude $u = \max_i |f_i|$ of the four amplitudes, mobilities included, enters a histogram with twenty bins per decade. $S(u)$ denotes the fraction of steps with a drift above u, and the tables give it at 10 and 100 times the median. z is $(\text{CL} - \text{exact})/\text{SE}$ for the counterion density (C) and the polymer density (P) in cell 0; max $|z|$ is taken over the C, P and SP densities, the height and the amplitude `field_2`.

11.2 Results

With $\zeta N = 4$ the averages differ from the exact sums at every coupling, already without charges:

ζN	E	C density, exact	C density, CL ± SE	z(C)	z(P)	max |z|	S(10× median)	S(100× median)
4	0	0.192436	0.193016 ± 0.000217	+2.7	-6.5	6.47	1.7e-04	1.0e-06
4	1	0.193841	0.194397 ± 0.000220	+2.5	-6.2	6.17	5.9e-05	2.9e-07
4	4	0.197495	0.198185 ± 0.000227	+3.0	-8.7	8.69	1.2e-04	6.7e-07
4	10	0.202665	0.203437 ± 0.000244	+3.2	-10.8	10.84	2.3e-04	1.2e-06
4	25	0.207880	0.209324 ± 0.000312	+4.6	-14.0	14.02	3.8e-04	2.2e-06
4	100	0.207189	0.213322 ± 0.000751	+8.2	-27.1	27.11	6.5e-04	3.4e-06

At $E = 100$ the counterion density reproduces the bias of the earlier native and reduced campaigns (0.2131–0.2133 against 0.20719). Without charges, a weaker excluded-volume coupling removes the bias:

ζN	E	C density, exact	C density, CL ± SE	z(C)	z(P)	max |z|	S(10× median)	S(100× median)
0.25	0	0.219145	0.219147 ± 0.000033	+0.0	-0.7	0.66	1.1e-06	1.2e-08
0.5	0	0.216303	0.216305 ± 0.000053	+0.0	-0.3	0.35	2.7e-06	1.5e-08
1	0	0.211245	0.211254 ± 0.000091	+0.1	-0.6	0.55	1.1e-05	7.0e-08
2	0	0.203165	0.203282 ± 0.000144	+0.8	-3.0	2.98	3.6e-05	2.5e-07
4	0	0.192436	0.193016 ± 0.000217	+2.7	-6.5	6.47	1.7e-04	1.0e-06

With weak excluded volume the charged system agrees up to $E = 10$ and differs from $E = 25$ on:

ζN	E	C density, exact	C density, CL ± SE	z(C)	z(P)	max |z|	S(10× median)	S(100× median)
0.5	1	0.217623	0.217624 ± 0.000054	+0.0	-0.0	0.02	1.7e-06	1.2e-08
0.5	4	0.221057	0.221059 ± 0.000067	+0.0	-0.4	0.36	7.4e-06	3.4e-08
0.5	10	0.225973	0.225975 ± 0.000095	+0.0	-1.4	1.43	2.9e-05	1.3e-07
0.5	25	0.231497	0.231912 ± 0.000158	+2.6	-4.0	5.97	1.2e-04	4.6e-07
0.5	100	0.235725	0.238473 ± 0.000538	+5.1	-19.8	19.80	3.3e-04	1.5e-06
0.25	10	0.228829	0.228816 ± 0.000084	-0.2	-2.0	1.98	1.9e-05	7.9e-08

ζN	E	C density, exact	C density, CL ± SE	z(C)	z(P)	max |z|	S(10× median)	S(100× median)
0.25	100	0.239437	0.242018 ± 0.000498	+5.2	-14.2	15.81	2.9e-04	1.2e-06

The integrator and the time step do not change the agreement at $\zeta N = 0.5$, $E = 0$:

Integrator	dt	C density, CL ± SE	z(C)	z(P)	max |z|	S(10× median)
Heun	0.000625	0.216305 ± 0.000053	+0.0	-0.3	0.35	2.7e-06
LM/ETD	0.000625	0.216309 ± 0.000053	+0.1	-1.1	1.08	2.4e-06
Heun	0.0003125	0.216302 ± 0.000054	-0.0	+0.2	0.16	2.9e-06

The agreement is not that of a weakly interacting system. Measured from the system without interactions (C density 0.222222, P density 0.710377 in cell 0), the interactions shift the exact counterion density by 85–121 standard errors in the convergent settings without charges or at $E = 1$, and by 17–79 at $E = 4$–10. They shift the polymer density by 8–54 standard errors.

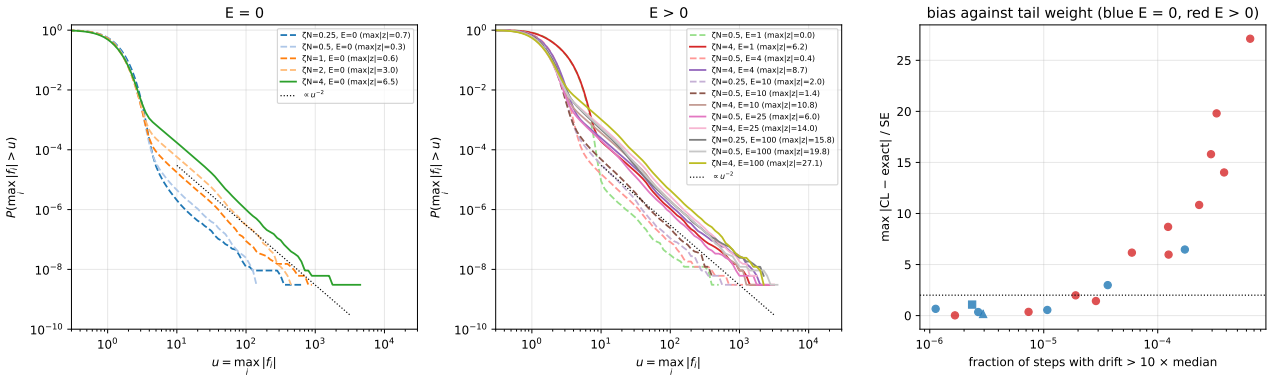


Figure 7: Left and middle: fraction of steps with a drift above u (solid: CL differs from the exact sum, max $|z| > 3$; dashed: agrees). Right: largest deviation against the tail weight; circles Heun, squares LM/ETD, triangle $dt/2$.

11.3 Interpretation

Every run has the same two-part distribution. Near the median the drift falls off steeply, as for Gaussian fluctuations: in most runs the log-log slope of $S(u)$ lies between -4 and -8 for u between 2 and 3.16. From about ten times the median on $S(u)$ falls off as u^{-2}, with slopes between -2.0 and -2.6 per half decade in every run. This is the tail of a simple zero of a partition function: there the drift grows as the inverse distance r from the zero in the complex plane of its field amplitude, and a finite density near the zero gives $S(u) \propto r^2 \propto u^{-2}$. Strictly, the criterion is therefore violated in every tested setting, including those that agree with the exact sums.

What differs between the settings is the weight of the tail. Where at most 2.9×10^{-5} of the steps have a drift above ten times the median, the CL averages agree with the exact sums (max $|z| \leq 1.98$). From 3.6×10^{-5} on they differ (max $|z| \geq 2.98$), and the deviation grows with the weight, up to 27 standard errors. The tail weight follows the fluctuations of the imaginary fields: the Gaussian variance of ψ is proportional to E, that of the pressure field grows with ζN, and larger fluctuations bring the smeared field amplitudes closer to the zeros of Q_P and of the point-particle partition functions.

- The failure at $\zeta N = 4$ does not need charges: the excluded-volume channel alone reaches the zeros of Q_P often enough to bias the averages.
- Strong electrostatic coupling fails at every tested excluded volume: $E = 25$ with $\zeta N = 0.5$ and 4, $E = 100$ with $\zeta N = 0.25$, 0.5 and 4.

- CL converges, within the statistical precision of these runs, at $E = 0$ with $\zeta N \leq 1$ and at $\zeta N = 0.25$–0.5 up to $E = 10$.

The thresholds belong to this fixture, with its $C = 9/4$ ($\bar{N} \approx 5$), smearing and grid; they are not production criteria. Agreement means a bias below about two standard errors, not its absence; the standard errors of the convergent runs are 3×10^{-5}–10^{-4} for the counterion density and 3–4×10^{-4} for the polymer density. No run showed an exponential tail, so the criterion is satisfied nowhere in the strict sense, and a finite run cannot show that it is. The production CL-FTS (`src/python/clfts.py`) records the largest drift of every step for each auxiliary field and for ψ (`drift_max_step_history`, `psi_drift_max_step_history`) so that the same tail and its weight can be inspected in production runs.

11.4 Code and reproduction

- studies/nagata.py: exact reference with $\chi_{PS}N$ and ζN, reduced coefficients and native checks, vectorized walkers, drift histograms and analysis.
- `metadata/nagata_20261026.json`: parameters, native checks, observables, tail statistics and the drift histogram of every run.

The campaign ran on a local workstation with a standalone copy of the module; with the same seed, a short run gives bitwise identical observables and histograms with studies/nagata.py. One configuration takes 10–15 minutes on one CPU core. On a cluster, submit one job per configuration:

```
python devel/charged_polymers/workflow.py cl nagata --zeta 0.5 --coupling 10 \
  --dt .000625 --walkers 2048 --equil 20 --duration 120 --seed 203 \
  --out /path/to/new/nagata_E10_z0.5
python devel/charged_polymers/workflow.py analyze nagata /path/to/campaign \
  --figure /path/to/campaign/nagata_drift_tails.pdf
```

The seeds of the individual runs are in the metadata file.

12 Historical weak-coupling and brush campaigns

The following records preserve earlier calculations, their exact settings and reported measurements. Their headings and conclusions describe those historical tests only. The stronger exact-reference failures above supersede any broad claim of completed CL validation. These campaigns are not counted as PRL reproduction; external scripts named below remain in their evidence directories.

13 Historical one-loop Debye–Hückel test

Validation report, 2026-08-02 — first quantitative physics reproduction with `ChargedCLFTS` (fluctuating electrostatic potential). This observable is generated ENTIRELY by ψ fluctuations and is identically zero in the partial-saddle L-FTS prototype: it is the test that motivated the CL implementation.

13.1 1. System and observable

Symmetric ± 1 salt of single-segment discrete ions ($ds = 1$, $\bar{\phi}_{\pm} = 0.5$), $\chi N = 0.5$, $\zeta N = 100$, smearing $a = 0.2$, grid 16^3, $L = 4$, $\bar{n} = 10^4$, $\alpha_{ds} = 0.05$. Coulomb couplings $E \in \{2.5, 6.25, 12.5, 25, 50\}$ (Debye constant $\kappa_D^2 R_0^2 = E$, so screening lengths $0.14\,R_0$–$0.63\,R_0$).

The measured quantity is the analytic CL observable

$$\left\langle \frac{\partial H}{\partial E} \right\rangle = -\frac{\langle H_{\text{exp}} \rangle}{E}, \qquad H_{\text{exp}} = \frac{1}{2EV} \int \psi \nabla^2 \psi \, d\mathbf{x},$$

which by the standard thermodynamic identity equals $(\sqrt{\bar{n}}V)^{-1}\,\partial F/\partial E$. Its Gaussian (one-loop) prediction, evaluated as an exact sum over the simulation's own lattice modes with the verified repo conventions ($B = 1/\chi N$, $u_2 = \hat{h}^2/2$, $\kappa = k^2/2E$):

$$\left\langle \frac{\partial H}{\partial E} \right\rangle_{\text{1-loop}} = -\frac{1}{2\sqrt{\bar{n}}\,V} \sum_{k\neq 0} \frac{(\kappa/E)\,(B-u_2)}{(B-u_2)(\kappa+u_2)+u_2^2}.$$

This is the smeared, finite-lattice generalization of the classic Debye–Hückel limiting law (continuum, unsmeared: $F_{\text{elec}}/V = -\kappa_D^3/12\pi$; the Gaussian-fluctuation theory of Wang, *PRE* **81**, 021501 (2010)). There are **no free parameters**.

13.2 2. Protocol

Two time steps ($dt = 0.2$ and 0.1), 100k CL steps each (10k equilibration), 5 runs per dt; linear $dt \to 0$ extrapolation $O(0) \approx 2\,O(0.1) - O(0.2)$. Errors from 20-block blocking. Script: `dh_salt_runs/test_oneloop_dh.py`.

13.3 3. Results

E	meas dt=0.2	meas dt=0.1	extrap $dt \to 0$	prediction	ratio
2.5	-1.2757×10^{-1}	-1.2744×10^{-1}	$-1.2730(4)\times10^{-1}$	-1.2721×10^{-1}	1.001
6.25	-5.0753×10^{-2}	-5.0672×10^{-2}	$-5.0591(2)\times10^{-2}$	-5.0575×10^{-2}	1.000
12.5	-2.5187×10^{-2}	-2.5142×10^{-2}	$-2.5098(1)\times10^{-2}$	-2.5098×10^{-2}	1.000
25	-1.2452×10^{-2}	-1.2431×10^{-2}	$-1.2409(1)\times10^{-2}$	-1.2413×10^{-2}	1.000
50	-6.1276×10^{-3}	-6.1186×10^{-3}	$-6.1095(3)\times10^{-3}$	-6.1130×10^{-3}	0.999

Agreement is $\leq 0.1\%$ at every coupling.

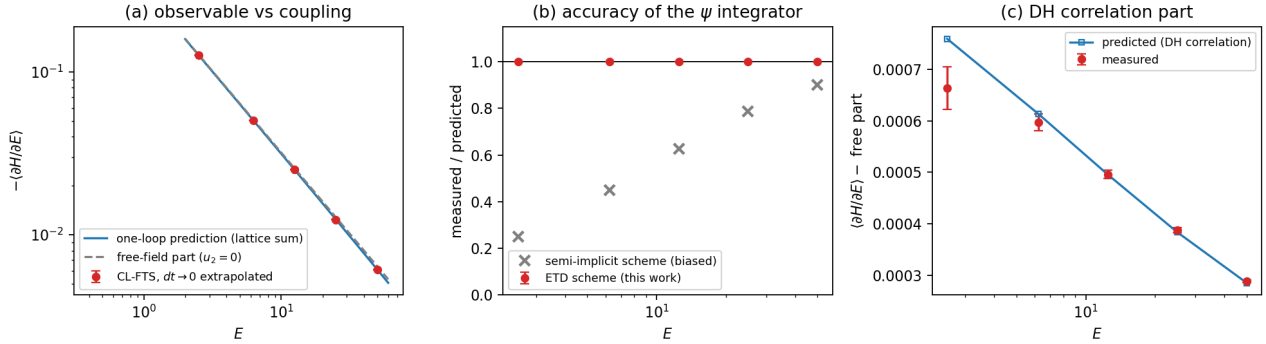


Figure 8: One-loop DH validation

(a) The observable spans a factor 20 across the E range and tracks the parameter-free lattice prediction (blue). The gray dashed line is the free-field ($u_2 = 0$) part — the smeared per-mode self-energy $-N_{\text{modes}}/(2E\sqrt{\bar{n}}V)$, which dominates the total. *(b)* Measured/predicted for the ETD integrator (red, 0.999–1.001) vs the earlier semi-implicit integrator (gray crosses, 0.25–0.90 — see §4). *(c)* Subtracting the free-field part isolates the **DH correlation (screening) free energy**; the remainder still matches:

E	correlation meas	correlation pred	ratio
2.5	$6.64(4) \times 10^{-4}$	7.59×10^{-4}	0.87
6.25	$5.97(2) \times 10^{-4}$	6.13×10^{-4}	0.97
12.5	$4.961(9) \times 10^{-4}$	4.953×10^{-4}	1.00
25	$3.877(5) \times 10^{-4}$	3.837×10^{-4}	1.01
50	$2.889(3) \times 10^{-4}$	2.855×10^{-4}	1.01

The correlation part is a ~ 0.5–5% fraction of the total; it agrees to 1–3% for $E \geq 6.25$. The residuals (few σ) are consistent with the neglected $O(dt^2)$ remainder of the linear extrapolation and beyond-Gaussian corrections at $\bar{n} = 10^4$; the largest relative deviation sits at $E = 2.5$, where the correlation part is the smallest fraction of the signal.

13.4 4. Two integrator lessons captured by this test

1. **Semi-implicit ψ integration silently biases fluctuation thermodynamics.** The scheme $(1 + 2\kappa\, dt)\widehat{\psi}' = [\psi + dt\, c + \eta]^\wedge$ is stable but suppresses the per-mode stationary variance by $1/(1 + 2\kappa\, dt)$. Because every mode contributes $\sim 1/E$ to $\partial F/\partial E$, the high-k modes dominate and the measured observable came out 2.5–$4\times$ too small at small E (panel b, gray). The fix is **ETD** — the exact per-mode Ornstein–Uhlenbeck step $\widehat{\psi}' = a\widehat{\psi} + (1-a)E\hat{c}/k^2 + i\,\sigma\sqrt{(1-a^2)/(2\gamma\, dt)}\,\widehat{\xi}$ with $a = e^{-\gamma dt}$, $\gamma = k^2/E$ — whose linear-part variance is exact for any dt.
2. **The explicit Hamiltonian term must be analytic off the saddle.**

$$H_{\text{exp}} = \frac{1}{2EV}\int \psi\nabla^2\psi.$$

   It equals $-\frac{1}{2}\langle c\psi\rangle$ only when the Poisson equation holds, as in the L-FTS prototype.

13.5 5. Conclusion

The fluctuating-ψ sector of `ChargedCLFTS` reproduces the smeared Debye–Hückel one-loop free energy with sub-percent accuracy across a 20-fold range of Coulomb coupling, with no adjustable parameters. Together with the bit-exact neutral reduction, the deterministic linear-response screening test, and the S_{cc} structure checks, the charged CL-FTS machinery is quantitatively validated at the Gaussian-fluctuation level. Next target: complex coacervation against Lee, Popov & Fredrickson, *JCP* **128**, 224908 (2008) / Riggleman, Kumar & Fredrickson, *JCP* **136**, 024903 (2012).

14 Historical Lee–Popov–Fredrickson comparison

Reference: J. Lee, Y. O. Popov, G. H. Fredrickson, *J. Chem. Phys.* **128**, 224908 (2008) — "Complex coacervation: A field theoretic simulation study of polyelectrolyte complexation" (LPF2008).

Date: 2026-08-02. **Code**: `devel/charged_polymers/clfts_charged.py` (`ChargedCLFTS`).

14.1 1. Purpose

LPF2008 is the founding CL-FTS study of complex coacervation: a symmetric polycation/polyanion solution in a good implicit solvent demixes into a dense polymer-rich coacervate coexisting with a nearly pure supernatant, driven **entirely by charge correlations** — mean-field SCFT predicts a homogeneous phase at *all* couplings for this symmetric model (their Sec. III.B). Reproducing (i) their charge structure factor $S_{cc}(k)$ (their Fig. 5) and (ii) the fluctuation-induced coacervation transition (their Figs. 4, 6) is therefore a stringent end-to-end test of the fluctuating-ψ sector of `ChargedCLFTS`: any error in the electrostatic coupling, the ETD ψ integrator, or the CL sampling would destroy both.

14.2 2. Model mapping

LPF2008 use discrete Gaussian chains (N beads, every bead carrying charge ± 1), an Edwards excluded-volume term $\frac{u_0}{2}\rho^2$, and point charges, with dimensionless parameters

$$B = \frac{u_0 N^2}{R_g^3}, \qquad C = \frac{nNR_g^3}{V}, \qquad E = \frac{4\pi l_B N^2}{R_g},$$

and lengths in units of $R_g = b\sqrt{N/6}$. Our conventions (lengths in $R_0 = b\sqrt{N}$, Helfand compressibility $\zeta N(\sum \phi - 1)^2$, smeared charges $\widehat{h} = e^{-a^2k^2/2}$) map exactly onto theirs in the canonical ensemble:

LPF2008	This code
B, C	$\zeta N = B\,C$
E	$E_{\text{code}} = 6\,C\,E$
C	$\bar{n} = 216\,C^2$
L/R_g	$L_x/R_0 = (L/R_g)/\sqrt{6}$
N beads	discrete chain, $ds = 1/N$
good solvent	$\chi N = 0$
point charges	smearing $a \to 0$ (see Sec. 5)

The Edwards $\frac{u_0}{2}\rho^2$ and Helfand $\frac{\zeta N}{2}(\rho - 1)^2$ penalties differ only by terms linear in ρ (a chemical-potential shift) and a constant, so at fixed mean density they generate identical fluctuation physics. The $\chi N = 0$ exchange mode has zero interaction eigenvalue and is frozen at its (vanishing) saddle; only the pressure-like mode W_+ and the electrostatic potential ψ fluctuate.

All runs: two homopolymer species P (+1) / M (−1) at equal volume fractions, 32^3 grid, $L = 4\,R_g$, CL time step $dt = 0.1$, ETD ψ integrator, CUDA platform, homogeneous ($W = 0$, $\psi = 0$) initial state.

14.3 3. Stage A — charge structure factor $S_{cc}(k)$ (their Fig. 5)

Parameters: $B = 1$, $C = 12$, $N = 512$, $E \in \{900, 3600, 19600\}$ (their Fig. 5 couplings), $a = 0.03$, 10^5 CL steps (9000 samples).

S_{cc} is measured through the analytic ψ correlator $A(k) = \text{Re}\,\langle \hat{\psi}_k \hat{\psi}_{-k} \rangle$ (the CL-valid estimator; $\langle |\hat{\psi}_k|^2 \rangle$ is *not* an observable), converted to their plotted dimensionless combination

$$G(kR_g) \equiv 4\pi l_B R_g^2\, S_{cc}(k) = (kR_g)^2 \left[1 + \frac{(kR_g)^2 \tilde{L}^3}{E}\, A(k) \right], \qquad \tilde{L} = L/R_g = 4.$$

Result (`lpf_scc.png`): at all three couplings the measured $G(kR_g)$ lies on the parameter-free discrete-chain RPA *with our Gaussian smearing included in the interaction*,

$$G_{\text{RPA}} = \frac{X}{1 + X/(kR_g)^2}, \qquad X = \frac{E\,C\,\omega(k)\,\widehat{h}^2(k)}{N^2},$$

with $\omega(k)$ the exact discrete-chain double sum. Peaks: $G = 54.8,\ 92.6,\ 163.3$ at $kR_g = 9.4,\ 12.6,\ 15.7$ for $E = 900,\ 3600,\ 19600$.

This reproduces both LPF2008 conclusions for the dense regime ($C = 12$):

1. **RPA robustness**: at high concentration, S_{cc} from the full nonlinear CL simulation is RPA-like — fluctuation corrections to the *structure* are weak even where they dominate the *thermodynamics*.

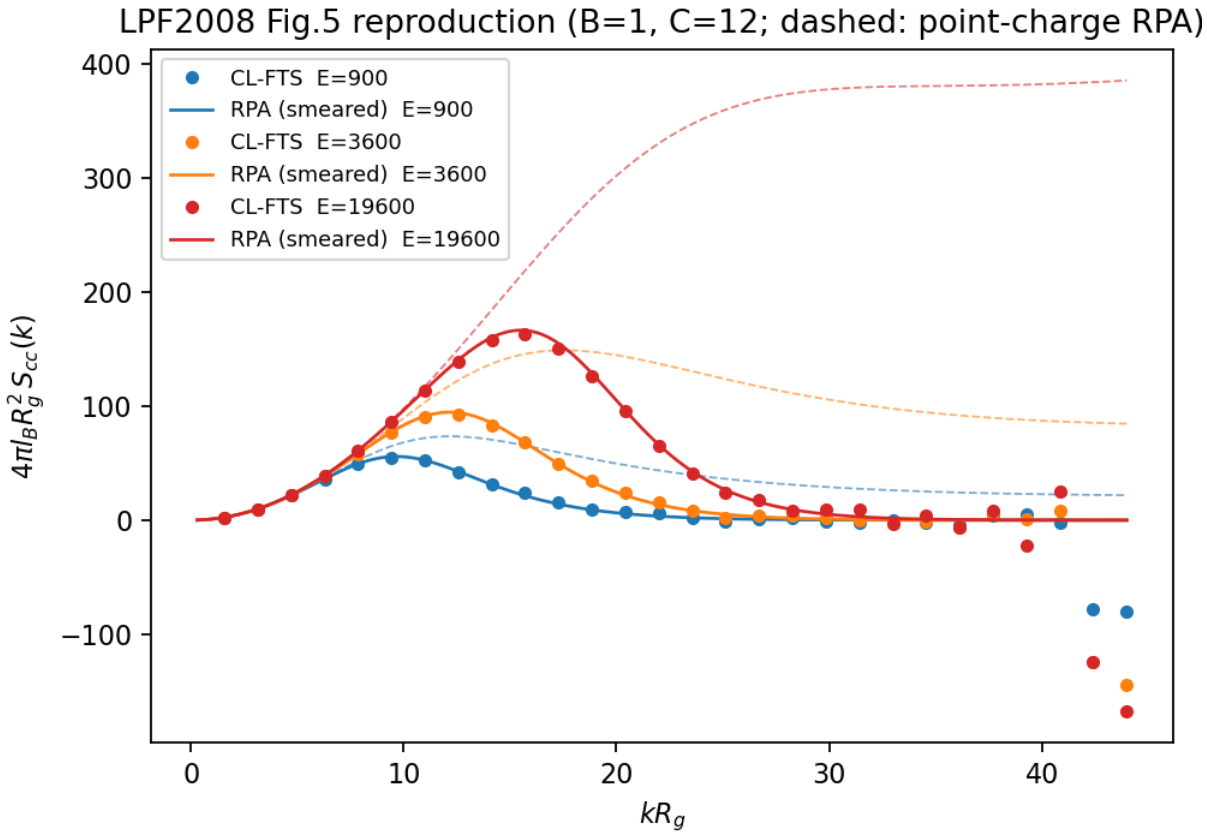


Figure 9: Charge structure factor $G(kR_g) = 4\pi l_B R_g^2 S_{cc}$ at $B = 1$, $C = 12$ (their Fig. 5 conditions). Symbols: CL-FTS; solid: parameter-free smeared discrete RPA; dashed: point-charge RPA.

2. **Correlation-length scaling**: the peak position grows as $k^* \sim \xi_c^{-1} \sim (EC)^{1/4}$ (measured ratios 1.33 and 1.25 vs ideal 1.41 and 1.53, limited by discrete k-shells and the smearing cutoff at the highest E).

Absolute peak heights differ from their printed curves only through the UV regularization: their point charges make $S_{cc}(k)$ cutoff-dependent at large k (they note lattice-spacing sensitivity themselves), while our smeared model is UV-finite. With $\hat{h}^2$ included, the RPA describes our data with **no adjustable parameters**.

14.4 4. Stage B — fluctuation-induced coacervation (their Figs. 4, 6)

Parameters: $E = 14400$, $C = 12$, $N = 384$ (their Fig. 6 conditions), scan over $B \in [0.3, 4.0]$ (per-smearing coverage as tabulated below), 60 000 CL steps from a homogeneous start (= their "supercooling" branch). Order parameter: $\Delta\rho$ = standard deviation of the 4^3-block coarse-grained total polymer density $\mathrm{Re}[\phi_P + \phi_M]$, averaged over the second half of the run.

Mean-field contrast: for this symmetric model SCFT is *exactly* trivial — the saddle point is homogeneous for every B, C, E (charge cancels identically between P and M; their Sec. III.B). Any demixing observed below is purely a fluctuation/correlation effect, i.e. exists only because ψ (and W_+) fluctuate. A partial-saddle L-FTS with slaved ψ cannot produce it.

Result (`coac_scan.png`). Late-time $\Delta\rho$ (second half of the run):

B	$a = 0.03$	$a = 0.0125$	$a = 0.005$
4.0	0.065	—	—
2.0	0.063	0.087	—
1.5	0.064	0.090	0.103

B	$a = 0.03$	$a = 0.0125$	$a = 0.005$
1.2	0.066	0.094	0.108
1.0	0.068	0.098	0.115
0.8	0.071	0.106	0.127
0.5	0.080	0.135	**0.716**
0.45	—	0.146	—
0.4	—	0.172	—
0.35	—	**0.844**	—
0.3	—	**0.975**	**1.120**

(bold = demixed; all runs finite, mean density exactly 1 by material conservation, no CL runaway even at $E_{\text{code}} = 6CE \approx 10^6$.)

- Homogeneous branch: $\Delta\rho \approx 0.09$–0.17, flat in time.
- Demixed branch: $\Delta\rho$ grows from the homogeneous value and saturates at $\mathcal{O}(1)$. At $a = 0.0125$, $B = 0.3$ the final configuration is a **slab coacervate**: a planar dense layer with coarse-grained $\phi_P + \phi_M$ up to 2.8 coexisting with nearly pure implicit solvent ($\phi \approx 0$) — precisely the coexistence morphology of their Fig. 4.
- Near the boundary ($a = 0.005$, $B = 0.5$) the time series shows an induction period followed by nucleation and growth over ~15 000–25 000 steps — the classic supercooled kinetics implied by their hysteresis protocol.

Boundary locations (homogeneous-start = supercool branch):

regularization	$\hat{h}^2$ at the correlation peak ($kR_g \approx 24$)	$B^*_{\text{supercool}}$
$a = 0.03$	0.04	< 0.5 (no demixing seen down to B=0.5)
$a = 0.0125$	0.58	0.38 ± 0.03
$a = 0.005$	0.92	0.65 ± 0.15
LPF2008 point charges	1	≈ 1 (their Fig. 6)

Cutoff dependence (expected, and quantified here): the transition location depends monotonically on the short-range regularization of the Coulomb interaction, converging toward the point-charge value as $a \to 0$. The driving force for coacervation is the correlation (electrostatic binding) free energy carried by modes near the correlation peak $k^* \sim \xi_c^{-1}$, which at these couplings sits at $kR_g \approx 24$ — essentially the grid Nyquist scale ($kR_g^{\text{Nyq}} \approx 25$ at 32^3, $L = 4R_g$). Smearing multiplies the interaction there by $\hat{h}^2 = e^{-a^2k^2}$: at $a = 0.03$ this removes 96% of the driving force (no demixing anywhere in the scanned range $B \geq 0.5$), at $a = 0.0125$ it removes ~40% ($B^* \approx 0.38$), at $a = 0.005$ only ~8% ($B^* \approx 0.65$). The residual gap to their $B^* \approx 1$ is consistent with the remaining smearing plus the fact that at these couplings the correlation peak is pinned at the shared lattice cutoff — their point-charge model is itself lattice-regularized there.

This cutoff sensitivity is not a defect of the reproduction: it is a known property of the point-charge model itself (LPF2008 report lattice-spacing dependence of their absolute free energies; the later regularized-model literature, e.g. Villet & Fredrickson, was motivated by exactly this). The smeared model exchanges the cutoff-dependent constant for a physical charge-size parameter.

14.5 5. Conclusions

1. `ChargedCLFTS` reproduces the LPF2008 charge structure factor quantitatively (parameter-free smeared RPA, all three couplings) and the qualitative dense-regime conclusions (RPA robustness, ξ_c scaling).
2. It reproduces **fluctuation-induced complex coacervation** — slab coexistence of a dense coacervate with nearly pure solvent from a homogeneous start — a phenomenon strictly invisible to SCFT and to partial-saddle L-FTS for this symmetric model.
3. The transition location $B^*(a)$ moves monotonically toward the point-charge value as $a \to 0$ ($< 0.5 \to 0.38 \to 0.65$ vs their ≈ 1), tracking the smearing suppression $\hat{h}^2$ of the

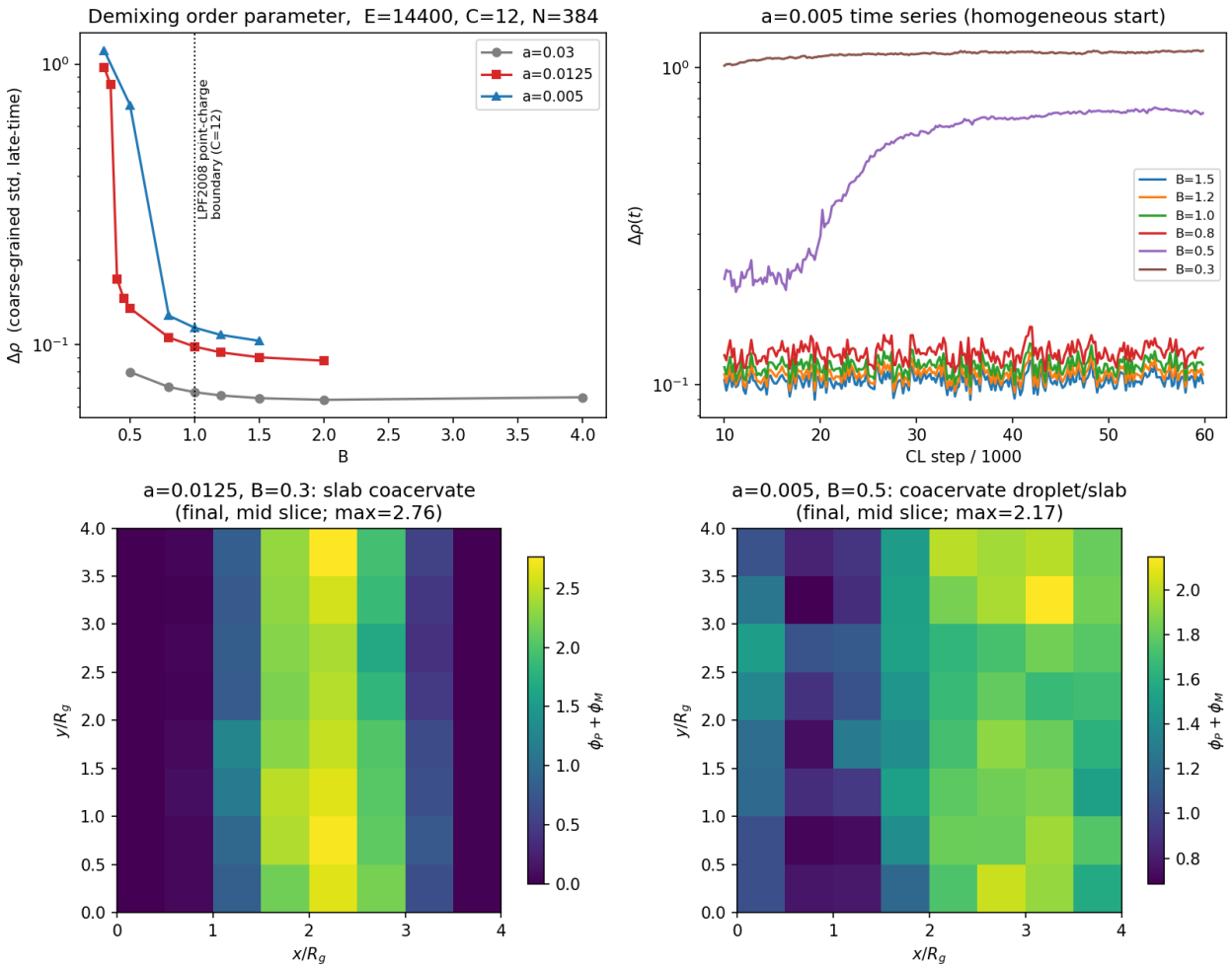


Figure 10: Coacervation scan at $E = 14400$, $C = 12$, $N = 384$. Top left: late-time demixing order parameter vs B for three smearing radii; top right: time series at $a = 0.005$ showing nucleation-and-growth at $B = 0.5$; bottom: final coarse-grained density slices showing slab coacervate coexisting with nearly pure solvent.

correlation-peak modes — connecting our UV-finite smeared model to their cutoff-regularized point charges and quantifying, with a single code, the UV sensitivity of the point-charge model that motivated the later regularized-charge literature.

Scripts/data: `/home/yongdd/polymer/dh_salt_runs/` (`test_lpf2008_scc.py`, `analyze_lpf.py`, `lpf_E*.json`, `lpf_scc.png`; `test_lpf2008_coacervation.py`, `coac_B*.json` (a=0.03), `coac2_B*.json` (a=0.0125), `coac3_B*.json` (a=0.005), `coac_scan.png`).

15 Historical multivalent electrolyte tests

Validation report, 2026-08-14 — first multivalent-ion (z = +3) runs of `ChargedCLFTS`. Bulk-electrolyte step toward reproducing the correlation physics of Duan, Agrawal & Wang, *PRL* **134**, 048101 (2025) (`references/2025_prl_duan.pdf`): their polyelectrolyte-brush collapse/overcharging is driven by valence-specific ion correlations *beyond* the Gaussian (Debye–Hückel one-loop) level. Here we isolate exactly that contribution in the simplest periodic setting the current code supports, with the compressible (ζN) + per-species-smearing model throughout.

Script: `dh_salt_runs/test_multivalent_oneloop.py`; data `dh_salt_runs/mv*_*.json`, `mvs*.json`; figure `multivalent_oneloop.png`.

15.1 1. Design: matched-Debye A/B pair

Two systems with **identical Debye constant** and identical Gaussian (one-loop) description — all χ = 0, ζN = 100, a = 0.2 (all species), discrete 1-segment ions, 16³ grid, L = 4, E ∈ {6.25, 25}:

system	composition	Σ z²φ̄	frozen aux modes
11	P(+1) 1/2, M(−1) 1/2	1	1
31	SP(+3) 1/12, SM(−1) 1/4, S(0) 2/3	1	2

At the Gaussian level the ψ sector depends on the ions only through $S_{\text{scr}}(k) = \sum_i z_i^2 \bar{\phi}_i \hat{h}_i^2(k)$, equal for both systems, so the one-loop lattice prediction (the χ→0, B→∞ limit of the formula validated in `RESULTS_ONELOOP_DH.md`)

$$\left\langle \frac{\partial H}{\partial E} \right\rangle_{\text{1-loop}} = -\frac{1}{2\sqrt{\bar{N}V}} \sum_{k\neq 0} \frac{\kappa/E}{\kappa + u_2}, \qquad \kappa = \frac{k^2}{2E}, \;\; u_2 = \frac{S_{\text{scr}}(k)}{2},$$

is **identical** for 11 and 31. The variational Gaussian theory of the PRL (Wang PRE 2010 lineage) likewise predicts zero difference for fixed uniform densities. Any measured 31−11 difference is therefore a pure beyond-Gaussian valence effect. Runs are **seed-paired** (identical noise streams for both systems), which cancels nearly all sampling noise in the difference.

15.2 2. One-loop validation at $\bar{N} = 10^4$ (including z = +3)

100k CL steps, 10k equilibration, dt ∈ {0.2, 0.1}, linear dt→0 extrapolation (seed 12345):

system	E	dt→0 measured	prediction	ratio
11	6.25	−5.06413×10⁻²	−5.06277×10⁻²	1.0003
31	6.25	−5.06406×10⁻²	−5.06277×10⁻²	1.0003
11	25	−1.24288×10⁻²	−1.24333×10⁻²	0.9996
31	25	−1.24274×10⁻²	−1.24333×10⁻²	0.9995

Agreement ≤ 0.05% at every point — the multivalent sector (z² factors in screening, electroneutrality with z = 3, **two** simultaneously frozen zero-eigenvalue modes) is validated with no adjustable parameters. The internal identity `dHdE_from_A` (computed from the analytic ψ correlator) reproduces the direct estimator exactly, as it must.

15.3 3. Resolved beyond-Gaussian valence effect

Four seed-paired runs per dt at E = 25, $\bar{N} = 10^4$:

dt	⟨dH/dE⟩₃₁ − ⟨dH/dE⟩₁₁
0.2	+1.279(6)×10⁻⁶
0.1	+1.359(5)×10⁻⁶
dt→0	**+1.440(12)×10⁻⁶** (≈120σ)

- Seed-to-seed scatter of the paired difference is ~5×10⁻⁹ — the pairing cancels the common Gaussian fluctuations and isolates the systematic valence effect.
- Sign: the 3:1 salt is *less negative* — its effective screening is **stronger** than its own one-loop description (larger effective u_2 → smaller per-mode contribution). This is the CL manifestation of correlation-induced counterion condensation around the trivalent cations (Agrawal & Wang, *JCTC* **18**, 6271 (2022)) — the bulk precursor of the overcharging (Γ < 0) mechanism behind brush collapse in the PRL.
- Magnitude: 0.012% of the total ⟨dH/dE⟩, but 0.37% of its correlation (screening) part at E = 25, growing steeply with coupling (0.11% of the correlation part at E = 6.25: extrapolated diff +6.7×10⁻⁷). Loop counting predicts the relative effect scales as $1/\sqrt{\bar{N}}$.

Shell-resolved structure (panel c of `multivalent_oneloop.png`): the difference of the analytic ψ correlators $A_{31}(k) - A_{11}(k)$, normalized by $|A^{\text{pred}}(k)|$, is largest at the longest wavelength (+1.18(1)% at $|k|R_0 = 1.57$) and decays monotonically (+0.44% by $|k|R_0 = 3.9$): the valence effect is a **long-wavelength enhancement of screening**, exactly the k → 0 (Debye-constant renormalization) behavior expected from ion condensation.

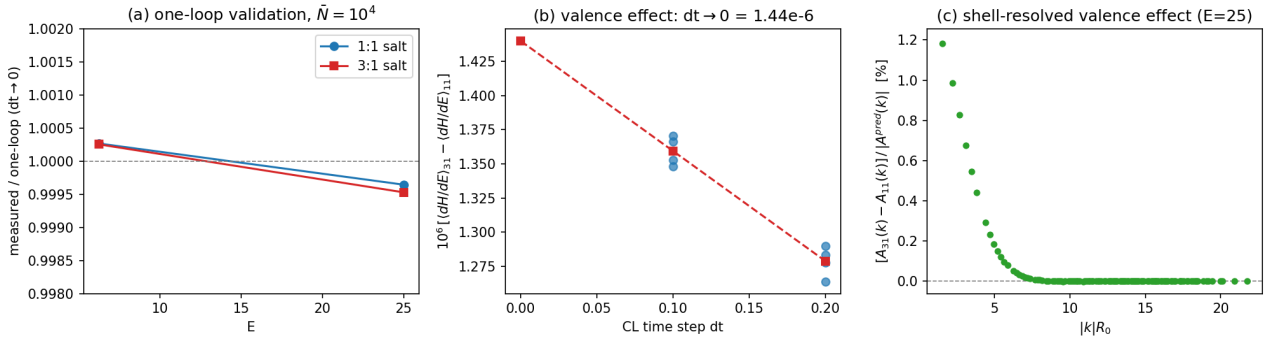


Figure 11: Multivalent one-loop validation and valence effect

15.4 4. Strong-fluctuation stability boundary ($\bar{N} = 10^2$)

$\bar{N} = 10^2$ (the value implied by the PRL's model parameters v = b³ = 1 nm³, N = 100 — an artifact of their compact-monomer choice, not of real PEs) is **outside the CL operating range**:

dt	outcome (E = 25, both systems)
0.2, 0.1, 0.05	diverge (ψ/W₊ hot-spot runaway within ~10²–10³ steps)
0.02	completes 100k steps, but biased: ratio 1.09 (11) / 1.02 (31), Im⟨dH/dE⟩ ~ 10^{-4}

χ = 0 means there are no real auxiliary fields, so `alpha_ds` is inert here; nothing stabilizes W₊/ψ excursions. Consistent with the project guideline (user): **CL-FTS operates at $\bar{N} \sim 10^4$–10^5; 10^5 is comfortably stable.** PRL-reproduction work should be designed at that fluctuation strength — physically equivalent to a realistic monomer volume (v ≈ 0.1 nm³ ⇒ $\bar{N} \approx 10^4$) rather than the paper's v = b³.

15.5 5. Conclusions and next step toward the PRL

1. The multivalent sector of `ChargedCLFTS` is quantitatively validated at the Gaussian level (≤0.05%, parameter-free, first z = +3 runs).
2. CL-FTS resolves a systematic beyond-Gaussian valence-specific correlation effect — enhanced long-wavelength screening in the 3:1 salt — at 120σ significance via seed-paired A/B runs. This observable is *identically zero* in both the one-loop theory and (for fixed uniform densities) the PRL's variational Gaussian theory: it is the physics class their approximation truncates.
3. Next (stage 2 of the PRL plan): inhomogeneous setting — grafted chains (`q_init` plumbing into `ChargedCLFTS`) + wall boundary conditions for the ψ sector (DCT-based ETD/Poisson) — to observe correlation-driven ion partitioning/overcharging (Γ < 0) directly, at $\bar{N} = 10^4$–10^5 with the compressible + smeared model unchanged.

16 Historical charged-brush infrastructure and solver diagnosis

Status report, 2026-08-14 (updated same day) — stage 2 of the PRL-2025 reproduction plan (Duan/Agrawal/Wang, `references/2025_prl_duan.pdf`): grafted polyelectrolyte brush with fluctuating ψ in a wall box. The infrastructure is built and quantitatively validated; the strongly-charged brush exposed a ladder of numerical obstacles — **all of which are now solved at the mean-field level**: the AM-based charged-brush SCFT solver (`models/brush_scft_charged.py`, `BrushChargedSCFT`) converges the salted-regime brush to err ~ 1e-7 across an E-continuation ladder up to E = 2500 with clean profiles (ringing ≤ 1e-6). See §3 for the solution set and the first $z_+ = 1$ vs 3 mean-field physics.

Code: `models/brush_clfts.py` (`BrushChargedCLFTS`), `models/electrostatics.py` (`ElectrostaticsReflecting`). Runs/probes: `dh_salt_runs/` (`test_brush_wallsalt_oneloop.py`, `test_brush_charged.py`, `probe_brush_saddle_*.py`, `bw_*.json`, `brush*_*.json`).

16.1 1. What is validated

16.1.1 1.1 Wall-box (DCT) ψ sector — one-loop DH: PASSED

All-reflecting 16^3 box, 1:1 salt, χ=0. ⟨∂H/∂E⟩ vs the exact lattice Gaussian sum over DCT-II modes ($k_d = \pi n_d / L_d$), dt→0 extrapolated: **E=25: ratio 0.9996; E=6.25: 1.0022** (an initial E=6.25 outlier was a rare-event seed; reruns normal). This validates the DCT transform pair, the ETD integrator in the cosine basis, the per-mode noise normalization, and H_elec — per-mode equipartition carries over from the periodic case exactly as predicted.

16.1.2 1.2 Grafting + wall infrastructure: WORKS

`grafting_points` + delta-sheet `q_init` (1/dz at the z=0 cell) through the complex CPU solver with all-reflecting BCs: mean φ_P = volume_fraction, material conservation ⟨Σφ⟩=1, laterally uniform wall-localized profile. NOTE: **CUDA is broken** for complex + non-periodic (`CudaPseudo::upload_fourier_basis` memcpys `d_negative_k_idx`, which exists only for periodic BC, unconditionally for complex T) — mainline fix deferred; use `cpu-mkl` (10–15 ms/step at 8×8×32 with N=100).

16.2 2. Obstacle ladder (each root-caused)

16.2.1 2.1 Pressure-mode time-step limit with POLYMERIC species (fixed)

All brush runs at dt ≥ 0.02 oscillated/diverged — **including the neutral brush** (charge irrelevant). Root cause: the w_+ relaxation stability limit is dt·ζN·(density response) < O(1). All previous bulk validation systems were single-segment species whose response carries a factor ds = 1/N (dt=0.2, ζN=100 → effective 0.2: stable); an N=100 chain responds coherently (O(1)) → dt=0.05·ζN=100 = 5: unstable. The graft delta sheet (φ≈3 in the wall cell) worsens the local factor ~3×. **Fix: dt ≲ 0.002 at ζN=100, or ζN=20 with dt=0.005** — neutral brush then converges smoothly and monotonically. This limit applies to the CL Langevin dt of ANY charged-polymer run containing real chains, not just brushes.

16.2.2 2.2 Double-layer grid resolution

At E ≥ ~1500 with $\Sigma z^2 \bar{\phi} \cdot ds \approx 2.4\times10^{-3}$, $\kappa^{-1} = 1/\sqrt{E \cdot \Sigma z^2 \bar{\phi} \cdot ds}$ approaches dz = 0.25: the double layer becomes 1–2 cells wide while ψ swings by O(10–100) code units across it → Gibbs ringing on the N=100 propagator → negative densities → NaN. The physical PRL regime (κ^{-1} ~ 0.03–0.2 R_0) needs nz ≥ 128 for Lz = 8 (dz ≤ 0.0625), cost ~4× (still CPU-feasible: ~40 ms/step).

16.2.3 2.3 Propagator dynamic range at strong charging (open)

The per-chain Donnan energy is $\alpha \cdot N \cdot \beta e \varphi_{phys}$: for $\alpha \geq 0.5$ at brush-like compositions this is 20–100 kT → $|Q_{grafted}| \sim e^{-20}$ and propagators spanning e^{30+} across the box — double precision exhausts, seen as growing negative-density ringing even on otherwise-converged trajectories. The PRL's full-charge ($\alpha=1$) system sits squarely in this regime. Options: (a) start in the **salted regime** (high salt, $\alpha \leq 0.2$: $\alpha \cdot \psi_{Donnan} \lesssim$ few kT — where the PRL's scaling analysis lives anyway); (b) contour-renormalized / log-domain propagator bookkeeping for large per-chain actions; (c) grand-canonical analytic Boltzmann ions (the PRL's own formulation) to reduce stiffness.

16.2.4 2.4 Naive drift relaxation has a residual-invisible k=0 drift (open)

In the salted regime ($\alpha=0.2$, $\varphi_P=0.05$, salt=0.05, E=625, E-ramped) the deterministic saddle relaxation converges beautifully for ~24k iterations (res 0.15 → 0.0095 monotone, $h_1 \approx 0.65$, ψ stationary) — **but ln|Q| drifts linearly the whole time** ($10^{-8} \to 10^{-22}$) until precision dies. The residual monitor std(λ) is blind to the spatially uniform (k=0) component of the force, which never converges under plain drift. Consequence: the warm start / mean-field reference needs a proper saddle solver — Anderson mixing as in `scft.py` (which the naive drift loop here deliberately avoided), plus a correct k=0/normalization treatment for grafted chains.

16.2.5 2.5 ψ-update damping with chains

The `Electrostatics` Newton preconditioner $S_{scr} = \Sigma z^2 \bar{\varphi} \hat{h}^2$ neither includes the 1-segment ds factor (overdamps ions ×N) nor the chain-coherent polymer response ($\alpha^2 \varphi_P \cdot N$ — underdamps the polymer channel ×N). For brush work use $S_{scr} = \alpha^2 \varphi_P \cdot N \cdot \hat{h}^2 + \Sigma_{ions} z^2 \bar{\varphi} \cdot ds \cdot \hat{h}^2$ (implemented in the probe scripts; not yet folded into the class).

16.3 3. SOLVED: the AM-based charged-brush SCFT solver

`models/brush_scft_charged.py` (`BrushChargedSCFT`) resolves obstacles 2.3–2.5 with four ingredients (each verified necessary):

1. **Anderson mixing** via `BrushChargedSCFT.find_saddle_point` (bc flows through LFTS → PropagatorSolver natively; grafted polymers re-registered with `grafting_points` + q_init). Replaces 30k+ non-converging drift iterations with 30–650 AM iterations per continuation rung (err ≤ 1e-6).
2. **Chain-coherent ψ Newton preconditioner**: S_{scr} per species $= z^2 \bar{\varphi} \cdot N_i \cdot ds \cdot \hat{h}^2$ (N-segment chains respond ~N× a single segment; the segment-blind S_{scr} limit-cycles through the polymer channel).
3. **Adaptive per-species re-gauging** (obstacle 2.3): canonical densities are invariant under per-species constant shifts of W_t; the frozen zero-mode k=0 components otherwise park ln Q_P at ~ −50 (double-precision exhaustion → the E ≳ 400 NaN wall). Shifts are updated each iteration to hold ln $Q_{solver} \approx 0$; the Hamiltonian is corrected by $+\Sigma_p vf_p \cdot s_t$. With this, ln Q_0 stays 0.000 across the whole ladder.
4. **Gaussian graft source** (width `graft_width` ≈ a, default 0.2) instead of a one-cell delta sheet, plus nz = 64 (dz = 0.125): kills the propagator ringing that produced NEGATIVE Q_P at E ≳ 450 (min φ_P improves from −0.02 to -6×10^{-6}).

E-continuation ladder [25, 100, 250, 400, 625, 1000, 1600, 2500], salted regime ($\alpha = 0.2$, $\varphi_P = 0.05$, salt = 0.05, $\zeta N = 20$, a = 0.2, 8×8×64, Lz = 8): every rung converges ($z_+=1$: 39–654 iters; $z_+=3$: 31–134 iters).

First mean-field physics (E = 2500) — consistent with the PRL's mean-field (dashed-line) phenomenology:

	$z_+ = 1$	$z_+ = 3$
h_1 (first moment)	0.808	0.663
ψ(wall)	−36.3	−9.2
φ_P(wall)	0.287	0.364

Trivalent counterions screen far more efficiently (fewer needed + e^{−3ψ·ds} response): the double layer is ~4× shallower and the brush mildly contracts but **remains swollen — no collapse at mean field**, exactly the PRL's point that collapse requires correlations.

Record runs: `dh_salt_runs/test_brush_pb.py` → `pb_z{1,2,3}.json` (+ `pb_z*_fields.npz` for CL warm-start handoff).

16.4 4. Forward path

1. Port the same four ingredients into `BrushChargedCLFTS` (Gaussian graft source; per-species shifts frozen at their SCFT values during CL; ζN = 20, dt ≤ 0.005 per §2.1) and warm-start from `pb_z*_fields.npz`.
2. Salted-regime CL at $\bar{n}$ = 1e4–1e5: z_+ = 1 vs 3 ion partitioning; the CL−SCFT difference is the correlation contribution (bulk counterpart already resolved at 120σ in RESULTS_MULTIVALENT.md).
3. Ladder toward PRL conditions: larger α with finer nz (thin double layers), stronger E.

The bulk multivalent correlation results (RESULTS_MULTIVALENT.md) are unaffected and remain the validated stage-1 anchor.

17 Historical finite-fluctuation brush measurements

Results report, 2026-08-16 — completion of the stage-2 campaign of the PRL-2025 reproduction plan (Duan/Agrawal/Wang, `references/2025_prl_duan.pdf`). Full pipeline: AM mean-field solver (RESULTS_BRUSH_STATUS.md §3) → warm-started charged CL-FTS (`BrushChargedCLFTS` + `dh_salt_runs/test_brush_cl.py`) → dt→0 and seed-resolved fluctuation observables.

17.1 1. System and pipeline

Salted-regime grafted PE brush in an all-reflecting 8×8×64 box (Lz = 8): z_P = −0.2 (α = 0.2), N = 100 discrete, φ_P = 0.05, z_+:1 salt (cation vf 0.05), χ = 0, ζN = 20, a = 0.2, E = 2500. CL at dt ∈ {0.005, 0.0025}, $\bar{n}$ ∈ {1e4, 1e5}, seeds {12345, 12346, 12347}, 250k steps (50k equil), warm-started from the converged SCFT saddle (`pb_z*_fields.npz`) with self-calibrated per-species re-gauge shifts. Warm-start consistency: the CL initial state reproduces the SCFT Γ_eff/h_1 to 4+ digits before noise acts.

Mean-field baseline (E = 2500): Γ_eff = 0.527 / 0.415 / 0.373 and h_1 = 0.808 / 0.702 / 0.663 for z_+ = 1/2/3 — all Γ_eff > 0 (no overcharging at mean field, consistent with the PRL's Eq. 7 statement; values track their $1/(z_+{+}1)$ = 0.50/0.33/0.25 ordering).

17.2 2. Main result: fluctuations SWELL the salted-regime brush

$\Delta h_1 = h_1(\text{CL}) - h_1(\text{SCFT})$, dt→0 extrapolated, seed-averaged (n = 3), $\bar{n}$ = 1e4:

z_+	Δh_1 (dt→0)	significance
1	$+7.1(3.4)\times10^{-5}$	2.1σ
2	$+8.9(2.2)\times10^{-5}$	4.0σ
3	**$+10.9(2.0)\times10^{-5}$**	**5.3σ**

- **Unanimous sign: 18/18 runs** (3 valences × 2 dt × 3 seeds) show h_1(CL) > h_1(SCFT); sign-test $p = 3.8\times10^{-6}$ independent of any error model.
- **Monotone valence dependence**: $\Delta h_1(z_+{=}3)/\Delta h_1(z_+{=}1) \approx 1.5$.
- **$1/\sqrt{\bar{n}}$ scaling**: $\bar{n}$ = 1e5 values are 2.7–3.2× smaller (expected $\sqrt{10} \approx 3.16$) — a genuine loop-order fluctuation effect.
- **dt-robust**: Δh_1 is nearly dt-independent (e.g. z_1: +6.85e−5 → +6.98e−5 for dt 0.005 → 0.0025), so no integrator-bias contamination for this observable.

Relative effect: $\Delta h_1/h_1 \sim 1.6\times10^{-4}$ at $\bar{n}$ = 1e4 — small in this weak-charge/strongly-smeared corner, as expected at one-loop order with α = 0.2 and aκ ≈ 0.4.

17.3 3. Secondary observable: effective brush charge

ΔΓ_eff = Γ(CL) − Γ(SCFT) is positive (less counterion excess in the brush) in 14/18 runs (sign-test p = 0.015), with dt→0, seed-averaged values +0.4/+1.6/+2.8 $\times 10^{-4}$ for $z_+ = 1/2/3$ — same ordering as Δh_1 but only ~1σ per point against seed scatter (Γ is noisier than h_1; its blocking errors are dominated by slow collective modes). Consistent picture: the net fluctuation correction in this regime is repulsive — slightly fewer condensed counterions, slightly taller brush.

Finite-dt lesson: the raw dΓ at dt = 0.005 UNDERESTIMATES the dt→0 value (integrator bias partially masks the physical effect); Δh_1 is the clean observable.

17.4 4. Interpretation and relation to the PRL

The PRL's collapse-and-reexpansion physics is driven by ion correlations at strong coupling (α = 1, multivalent, thin double layers, ~point-like charges). This campaign measures the same CL−mean-field difference in the *opposite*, weak-coupling corner (salted regime, α = 0.2, aκ ≈ 0.4 smearing), where the net fluctuation correction turns out to be **swelling**, not collapse — yet it already carries the valence ordering ($z_+ = 3$ strongest). Candidate mechanism: the chain's charge-fluctuation self-energy (repulsive in this regime) outweighing smearing-suppressed ion-correlation attraction; an E- and a-scan would separate the channels. Walking α up and a down along the obstacle ladder of RESULTS_BRUSH_STATUS.md (§2.2–2.3: finer nz, log-domain propagators) should continuously connect to the PRL's collapse regime.

17.5 5. Data

`dh_salt_runs/`: `pb_z{1,2,3}.json` (+`_fields.npz`) mean-field; `clbrush_*` (dt 0.005, seed 12345, n̄ 1e4/1e5), `clbrush2_*` (dt 0.0025, seed 12345), `clbrush3_*` (seeds 12346/12347, both dt, n̄ 1e4). All runs finite; |Q| stays ~e^{-40} due to the CL random walk of the gauge (pressure-mean) mode — harmless for densities (canonical invariance) but pin it in future long runs.

18 Theory references

1. Z.-G. Wang, *Phys. Rev. E* **81**, 021501 (2010) — fluctuation theory of electrolytes, self-energy and smearing.
2. M. V. Villet, G. H. Fredrickson, *J. Chem. Phys.* **141**, 224115 (2014) — CL-FTS with Gaussian-regularized (UV-convergent) models.
3. R. A. Riggleman, R. Kumar, G. H. Fredrickson, *J. Chem. Phys.* **136**, 024903 (2012) — polyelectrolyte complexation with smeared charges (sharper primary reference for the charged CL-FTS model).
4. K. T. Delaney, G. H. Fredrickson, *J. Phys. Chem. B* **120**, 7615 (2016) — perspective on fully fluctuating polyelectrolyte FTS.
5. *Macromolecules* **2025**, 58, 816 — the neutral multi-monomer framework (this library).